Labor Market Dynamics after Cost-of-Living Shocks*

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Abstract

This paper provides causal microevidence on how idiosyncratic cost-of-living shocks are transmitted to labor market outcomes. Using the case of energy price changes, I combine 20 years of German employer-employee data with a representative household expenditure survey and rely on regional differences in energy expenditure patterns to identify the pass-through of energy cost shocks into earnings. A one-standard-deviation shock raises annual earnings growth by 0.4 percentage points, offsetting 30-35% of local cost increases within a year and compensating for 80% over five years. Job-to-job mobility is a key mechanism: higher exposure to energy price changes both increases the likelihood of switching and makes switches more profitable. Through the lens of an equilibrium model of the labor market, I clarify that while labor market responses substantially mitigate the distributional impact of relative price changes, they impose additional non-pecuniary costs on workers.

Keywords: labor supply dynamics, heterogeneous inflation rates, energy price shocks

JEL classification numbers: J3, J62, E31

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1 Introduction

When relative prices change, households with different consumption baskets face different inflation rates. The surge in food and energy prices during the COVID-19 pandemic and following the war in Ukraine has brought renewed attention to this distributional concern, as necessities like energy constitute a larger share of poorer households' expenditures.[1] More generally, a growing literature documents that heterogeneity in expenditure composition is pervasive and creates differences in the real cost of living across households even when aggregate inflation is modest (e.g., Jaravel, 2021, 2024).

While significant advances have been made in understanding the degree of heterogeneity in personal inflation rates, we know comparatively less about how individuals respond. For instance, recent macroeconomic work suggests that when faced with increased living costs, workers have stronger incentives to renegotiate their wages with their current employer or to search for higher-paying jobs (Afrouzi et al., 2026; Bostanci, Koru and Villalvazo, 2025; Pilossoph and Ryngaert, 2025). However, we have little empirical evidence from microdata to guide the calibration of labor supply elasticities or earnings responses, and, in particular, we lack causal evidence on the link between individuals' exposure to price changes and corresponding labor market outcomes. If such labor-market adjustments are successful, heterogeneous inflation need not fully translate into heterogeneous losses in real consumption. Understanding the empirical extent of endogenous worker responses is thus highly informative for designing effective policy measures during periods of inflation.

The main contribution of this paper is to quantify the relevance of workers' labor market responses to changes in their cost of living. Answering this question poses the challenge that both labor supply decisions and price changes are potentially endogenous to local economic conditions. To address this issue, I focus on the case of energy prices, which are a key driver of inflation inequality between consumers (Corsello and Riggi, 2023), and exploit spatial variation in energy consumption patterns across German administrative counties. Using a representative household expenditure survey, I first document substantial cross-county heterogeneity in energy expenditure shares. Conditional on household characteristics, energy shares range from 4.6% of total spending in some counties to about 9.8% in others. This variation reflects differences

[1] In line with this, a series of recent studies highlight the role of consumption heterogeneity in shaping welfare implications of energy price increases (Kuhn, Kehrig and Ziebarth, 2021; Känzig, 2023; Oni, 2023; Pieroni, 2023; Gnocato, 2025) and other inflationary episodes (Cardoso et al., 2022; Del Canto et al., 2025; Pallotti et al., 2024).

in commuting infrastructure and housing stock rather than household sorting. Combined with common energy price changes, the county-specific expenditure shares yield a local energy-specific cost-of-living index that is structurally similar to a shift-share instrument. I then merge the cost-of-living index with 20 years of German-linked employer-employee data to investigate how cost shocks are transmitted to earnings and employment decisions.

A one-standard-deviation energy cost shock, equivalent to a 1.2% increase in the local cost of living, raises annual earnings growth by 0.4 percentage points. That is, adjustments in nominal earnings offset approximately one-third of real earnings losses within one year. While this still implies a divergence in real consumption between high- and low-exposure areas, the effect is reduced relative to a benchmark in which earnings adjust according to average inflation rates. Since it is natural to assume that not all labor market adjustments occur in the short term - e.g., due to mobility frictions or a lower frequency of wage adjustments - I also consider changes over two- to five-year windows. Over longer horizons, pass-through increases, with workers recovering about 57% of their losses over two years and 80% over five years. This further narrows the gap but does not fully close it across local labor markets.

The similarity between the local cost-index and shift-share instruments greatly facilitates the exposition of the necessary assumptions to interpret these estimates as causal effects. One must either rule out other shocks that correlate with energy prices or account for unobserved local conditions that mediate the impact of energy price changes and correlated shocks. I demonstrate that the results are unaffected when instrumenting energy price changes with carbon policy surprises (Känzig, 2023) and when allowing yearly fixed effects to vary over several local labor market indicators. Importantly, I compare only labor markets that are similar in industry composition, unemployment rates, and employment concentration, and I match workers based on their firms' 3-digit sector and size. This allows me to contrast two individuals who differ in their consumption profiles but work in similar labor markets and for firms that face comparable cost shocks from energy inputs. The effects thus capture differences in workers' labor-supply responses while adjusting for common changes in labor demand.

When thinking about how workers can recover part of their real earnings losses, a back-of-the-envelope calculation suggests that about two-thirds of the average response is attributable to changes in earnings among workers who remain with their current employer. The remaining third is due to selected job-to-job mobility. I find no evidence that, for job stayers, the increase in earnings is associated with switching from part-time to full-time employment, working overtime, being

promoted, or receiving bonus payments, indicating a role for on-the-job bargaining. In line with this interpretation, the pass-through of energy cost shocks into earnings is higher among workers who are plausibly better able to solicit and leverage credible outside offers, such as college-educated workers and younger workers.

Selected job-to-job mobility more generally plays a vital role in the adjustment process. On the one hand, workers in counties with higher exposure are more likely to switch employers when energy prices rise — a one-standard-deviation cost shock is associated with a 0.3 percentage point increase in the yearly rate of job mobility. On the other hand, conditional on switching, job changers in high-exposure areas also receive substantially larger wage increases than their counterparts in low-exposure areas. I provide evidence that this is partly due to targeting firms with higher overall earnings. The probability of observing job changes involving a reduction in earnings correspondingly declines during periods of greater energy-cost shocks.

The documented average nominal earnings responses produce a substantially muted change in real earnings inequality across markets. However, there are at least two reasons why this does not imply that the cost shocks are unproblematic from a welfare perspective. First, adjustments are not immediate and remain incomplete even over five-year windows. Inequality in real living standards is thus rising after energy price increases, despite labor-market adjustments. Second, the fact that workers are more willing to switch to higher-paying firms only after increases in their cost of living implies that such switches are not costless. To quantify welfare dispersion more comprehensively, I develop and estimate an equilibrium model of the labor market in which consumers choose among firms not only based on posted wages but also on their preferences for firms' non-pecuniary attributes.

The model builds on recent work that derives imperfect competition in the labor market from workers' heterogeneous preferences for firms' non-wage attributes (e.g., Card et al., 2018). Preference heterogeneity implies that firms are imperfect substitutes from the perspective of workers, thereby endowing them with a degree of wage-setting power. A key feature of my model is that workers reside in different labor markets characterized by location-specific energy subsistence requirements that capture differences in commuting needs and housing efficiency across areas. When energy prices rise, workers are poorer in real terms. This increases their marginal utility of income and, consequently, makes them more willing to sacrifice non-wage attributes of the firm for higher wages. The model provides a rationale for why, when costs are low, workers are willing to forgo higher earnings and even accept pay cuts when changing employers,

but target better-paying firms when costs are high. At the same time, firms internalize this elevated focus on wages as an increase in the elasticity of labor supply, which induces them to post higher wages throughout. The degree of adjustments in nominal earnings is more pronounced in labor markets with higher exposure to energy price changes.

I estimate the model's parameters by targeting both standard labor-market moments, such as the mean and variance of wages and the firm-size distribution, as well as the reduced-form pass-through coefficients from my empirical analysis. The estimation reveals heterogeneity in workers' preferences for firms' non-wage attributes and the relative importance they place on consumption. The inferred distribution of tastes implies that non-pecuniary welfare costs account for almost one-third of the additional welfare costs experienced by workers in high- relative to low-exposure labor markets. Energy cost shocks thus impose larger hidden welfare losses on already disadvantaged workers. Put differently, even when earnings almost fully adjust to cost shocks, workers are not fully insured because the adjustment process itself is costly. Related observations were recently made by Guerreiro et al. (2024) and Afrouzi et al. (2026), who both argue that workers' responses to inflation, such as bargaining, strikes, or job search, have welfare consequences beyond their impact on earnings.

Related Literature and Contribution. This paper connects and contributes to several strands of the literature in applied labor and macroeconomics. Recent empirical work indicates that wages fall less after energy price increases for individuals with less than a high school diploma than for their peers with higher levels of education (Del Canto et al., 2025). While the overall negative impact on wages likely reflects a decline in labor demand, the differentially weaker effect on less-educated individuals is consistent with this paper's results, as energy expenditure shares decline with education.[2] At the same time, Broer, Kramer and Mitman (2024) and Känzig (2023) provide evidence that after energy prices increase, wages fall more strongly at the bottom of the income distribution, whereas, in the case of changes in the aggregate price level, Afrouzi et al. (2026) and Blanco, Drenik and Zaratiegui (2024) find a slower real wage decline and a faster recovery at the bottom. In contrast to these previous studies, this paper focuses on workers' labor-supply responses to cost shocks, accounting for differences in labor demand. Combining household consumption data with high-quality labor-market registry data, I provide causal evidence on the

[2] Kehrig and Ziebarth (2017) study a complementary case to this paper, in which consumers are identical in their consumption bundles, but firms differ in the energy intensity of their production. Similarly, Mertens, Müller and Neuschäffer (2025) use firms' energy intensity to identify rent-sharing between workers and firms.

link between consumption baskets and earnings adjustments and calculate the share of real wage losses recovered through changes in earnings.

Closest to this paper, Ahlvik et al. (2025) use variation in the expiration dates of households' electricity contracts in Finland to study a variety of household-level adjustments during the European energy crisis of 2022/2023. They find that lower-income households, in particular, increased their labor supply and earnings. Relative to their study, I focus on fewer adjustment margins but provide an in-depth analysis of labor market dynamics. The linked employer-employee data that underlie my study allows the tracking of individuals over a 20-year period. I provide a detailed study of mechanisms such as on-the-job compensation, job-to-job mobility, and responses at the extensive margin of labor force participation.

The documented increase in selected job-to-job mobility connects to a long-standing literature that tries to understand why workers do or do not switch jobs, giving rise to classic work on job ladder dynamics (e.g., Topel and Ward, 1992; Moscarini and Postel-Vinay, 2016) and the idea of job mobility as a precautionary mechanism against uninsurable income risks (e.g., Attanasio, Low and Sánchez-Marcos, 2005; Blundell, Pistaferri and Preston, 2008; Low, 2005). The results of this study show that workers also respond to cost shocks by increasing labor mobility and are more likely to direct their job transitions towards higher-paying firms.

Darougheh et al. (2025) show that an increase in the intensity of job transitions feeds back into the price setting of firms and consequently inflation. Similarly to their paper, the results of my study highlight that inflation and labor market dynamics are intertwined, relating to the debate on price-wage spirals (see also, Faccini and Melosi, 2025; Moscarini and Postel-Vinay, 2017, 2023). While the majority of work in this area studies the path from wage growth to inflation, a few recent papers focus on the less-studied reverse path and investigate the impact of realized (Bostanci, Koru and Villalvazo, 2025) and expected (Pilossoph and Ryngaert, 2025) inflation on wages. They first show empirically that search effort increases in response to price shocks, and then use this fact to inform search models that predict a partial escape from real wage losses for employees by switching employers. Afrouzi et al. (2026) calibrate a large frictional labor market model with costly renegotiations and search effort to highlight that even if workers can recover real wage losses after price increases through job transitions, this comes at a utility cost that decreases welfare overall.

I contribute to this work in two important dimensions. First and foremost, the results of this paper make empirical progress by exploiting large-scale linked employer-employee data rather than surveys or aggregate data. In combination with a quasi-experimental framework based on

cross-sectional consumption heterogeneity, this allows me to identify causal impacts of cost shocks on the distribution of real wages for a representative set of workers. I provide estimates of earnings and job mobility elasticities to guide future work. Second, this paper's conceptual focus is on idiosyncratic rather than average inflation rates. In contrast to previous work that models the dynamics by which nominal earnings keep pace with the general price level, my focus is on the distributional impact of relative price changes. Even if consumers keep real earnings constant relative to average inflation, consumption heterogeneity, combined with relative price changes, can imply sustained real earnings losses for consumers who spend a larger share on hard-to-substitute goods (necessities). In fact, Dustmann, Fitzenberger and Zimmermann (2022) highlight that not accounting for differential trends in housing expenditures leads to an underestimate of the increase in income inequality in Germany between the 1990s and 2013.

Finally, and in line with the focus on inflation heterogeneity, the structural part of this paper embeds a consumption problem with cross-labor-market differences in minimum energy consumption levels within an otherwise standard wage-setting model of heterogeneous firms. The paper thus draws heavily from Card et al. (2018), Berger, Herkenhoff and Mongey (2022), Chan et al. (2024), Lamadon, Mogstad and Setzler (2022), and Volpe (2024), who rely on preference heterogeneity for amenities to derive insights on firms' market power and worker-firm rent-sharing. While not the primary focus of this paper, the model results provide estimates of labor market power in Germany that exploit variation in workers' costs of living rather than firm-level revenue shocks for identification. The key idea is that firms' equilibrium wage-posting responses result from a change in workers' marginal utility of income, which in turn endogenously alters their labor-supply elasticity. Mongey and Waugh (2025) apply a similar idea to the pricing decisions of firms and provide an in-depth discussion of how exogenous shifts in the marginal utility of income affect output prices and mark-ups.

Bloesch, Lee and Weber (2024) consider a model that combines on-the-job search with wage-posting firms. They argue that, in equilibrium, the role of on-the-job search in explaining the pass-through of cost-of-living shocks into wages is quantitatively limited because an increase in overall search effort also increases the probability that firms can find a replacement for any given worker. However, they consider a setting with ex ante identical firms and a symmetric equilibrium, in which all firms post the same wage. This implies that the only credible outside threat workers have is a quit into unemployment. I demonstrate that when firms differ in wages and non-pecuniary attributes, workers will rerank firms in response to cost shocks, triggering

equilibrium wage responses that align with my empirical estimates.

Road Map. The remainder of this paper is organized as follows: Section 2 introduces the institutional setting in which wages in Germany are set, as well as the data sources underlying the empirical analysis. Sections 3 and 4 present the empirical framework and results, respectively. Section 5 introduces a theoretical framework rationalizing the link between idiosyncratic costs of living and changes in labor supply decisions. Section 6 finally offers some concluding remarks.

2 Background and Data

Before turning to the empirical analysis, this chapter briefly outlines some key features of the German context in which the study is situated. It then describes the underlying data sources and discusses the measurement of key concepts. The link between heterogeneous consumption patterns, price trends, and labor market outcomes, which is necessary for the empirical exercise, naturally imposes a heavy burden on data requirements. Since no comprehensive data set containing all relevant information exists for Germany, I use and combine several data sources.

Background: Spatial Heterogeneity and Wage Setting in Germany

The empirical exercise relies on spatial heterogeneity to construct differences in the exposure to energy price shocks. In particular, differences in the share of total expenditures dedicated to energy across German counties. Today, Germany is divided into 16 regions, which together comprise 400 administrative counties. Counties include 106 independent cities and 294 further counties, varying vastly in size, population, and geography. The delineation of counties changed slightly over time: some were split to form newer, smaller counties, while others were unified. All data sets used in this study rely on a time-consistent geographic definition that includes 400 counties, which serve as proxies for local labor markets.

The goal is to relate the heterogeneity in the intensity of cost shocks across counties to changes in labor market outcomes. Since wages in Germany were traditionally determined by collective bargaining agreements at the more aggregate sector-by-region level, this link is not immediately evident. However, as described in great detail by Dustmann et al. (2014) and Jäger, Noy and Schoefer (2022), the German institutional arrangements for wage determination have undergone a drastic transition from a system primarily characterized by collective bargaining

agreements to one in which firm-level negotiations play an increasingly important role. Firm-specific work councils are a central player in this co-determination paradigm. Negotiations between them and individual employers offer greater flexibility relative to local conditions, increasing the likelihood of both upward and downward deviations from collectively bargained wages. Jung and Schnabel (2011) show that the payment of wage cushions on top of collectively bargained wage floors is common in Germany, and Boeri et al. (2021) illustrate that the downward flexibility of wages allows for a higher correlation between local productivity levels and wages in Germany than in Italy, which leads to lower geographical differences in unemployment.[3] More recently, Caldwell, Haegele and Heining (2024) provide evidence from firm and worker surveys that individual-level bargaining is an increasingly relevant feature of the German labor market. The degree to which this flexibility allows workers to be compensated for local increases in the cost of living is the empirical question this study seeks to answer.

Labor Market Data: SIEED and BHP by IAB-FDZ

For labor market outcomes, I mainly rely on the Sample of Integrated Employer-Employee Data (SIEED) provided by the research data center of the Federal Employment Agency at the Institute for Employment Research (IAB-FDZ). The basis for the SIEED data set is a 1% random sample of establishments in the Social Security registry, i.e., of all establishments with at least one employee liable to social security.[4,5] For each included firm and year, the SIEED links information on employment spells of all employees covering the period from 1975 (1991 for East Germany) to 2023. It further reconstructs their full employment histories from Social Security records. This also includes periods of work at firms not included in the 1% random sample. For them, firm-level information is linked from the Establishment History Panel (BHP, see below). The SIEED data is organized in employment spells, detailing the exact start day and month of each employment episode. I construct an individual-level yearly panel by classifying the employment spell an individual was in on December 31st of each year as the relevant job. If an individual was not employed on December 31st but was employed at another point in the year, I instead rely on this information. If an

[3] As highlighted by Card and Cardoso (2022), the payment of wage cushions on top of wage floors that resulted from collective bargaining agreements is, of course, a common occurrence in Europe that is not necessarily linked to the presence of worker councils alone.

[4] As of 1999, establishments with at least one marginal part-time employee are also included, irrespective of whether they also employ workers with contracts subject to social security.

[5] In the data, establishments are defined as "regionally and economically delimited units" (Ganzer et al., 2022), which is not equivalent to a production plant, since it aggregates plants within an administrative district/county. Although technically, a firm is, in some cases, a further aggregation of different establishments, the remainder of the text will refer to each observation in the BHP and SIEED data as a firm or an establishment interchangeably.

Table 1: Summary Statistics – SIEED Data

	Mean	*S.D.*		*Mean*
Daily Gross Income	105.82	60.22	Sector	
Age	40.85	8.21	*Agriculture & Fishing*	0.01
Female	0.28		*Mining & Quarrying*	< 0.01
German Citizen	0.93		*Manufacturing*	0.28
Residence: Eastern Germany	0.19		*Utilities*	0.01
Education			*Wholesale & Retail*	0.15
Below Abitur	0.70		*Hotels & Restaurants*	0.02
Abitur	0.13		*Transportation*	0.08
Academic Degree	0.17		*Financial Intermediation*	0.03
Job Task			*Real Estate & Renting*	0.16
Unskilled/Semiskilled	0.07		*Education*	0.02
Skilled	0.68		*Health & Social Work*	0.07
Complex Task	0.12		*Community & Other Services*	0.04
Highly Complex Task	0.13		*Public Administration*	0.04
Changed Job in Past Year	0.16		*Construction*	0.08
Changed Residence in Past Year	0.04		*Other*	< 0.01

Notes: The table displays summary statistics for the SIEED data, excluding individuals who are not in social security employment, commute to work from outside of Germany, or are not between 25 and 55 years old. Daily gross income is imputed using the procedure described in Dauth and Eppelsheimer (2020). *Abitur* is a dummy variable equal to 1 in case the individual obtained a university-qualifying high school degree. *Below Abitur* consequently indicates individuals with either no high school degree or a high school degree that does not grant access to universities.

individual has multiple employment spells simultaneously, I prioritize full-time over part-time contracts, then use the length of employment and daily income as further tiebreakers, retaining only the longer, higher-paid spell.

For each included employee, the SIEED contains socio-demographic information (age, gender, nationality, educational background) and allows the tracking of daily earnings and job-to-job transitions over time. Earnings are recorded as gross compensation over the complete employment spell, divided by the number of days. Given that the data source is the Social Security Registry, earnings are top-coded to the maximum contribution limit. This concerns 8.2% of all individual-year observations, and I impute top-coded earnings following the procedure in Dauth and Eppelsheimer (2020) to improve precision when exploring pass-through heterogeneity at the upper end of the income distribution.

I restrict the main analysis sample to the period from 1999 to 2019 and to prime-age workers (25–55 years) in full-time social security employment who reside in Germany ($\approx$ 0.4% commute from abroad). Information on the county of residence of each employee is only available from 1999 onward, placing restrictions on the earliest observations for which energy cost shocks can be constructed. Excluding observations from 2020 onwards avoids issues of non-standard labor market dynamics associated with the COVID-19 pandemic.[6] As a further sample selection criterion, I keep only workers with at least one spell of 5 consecutive employment years. Since I will be comparing labor market dynamics over up to 5-year windows, this ensures that changes in sample composition do not drive dynamic effects, but comes at the cost of selecting workers on labor force attachment. Note, however, that the definition of attachment is comparatively mild: any worker-year is considered as long as there is some form of employment during the year, irrespective of duration. Unemployment periods of less than a year, thus, do not lead to a sample exclusion. In total, the analysis sample for this study comprises 7,115,473 individual-year observations and 639,034 unique employees. Table 1 contains summary statistics for the socio-demographic makeup of the sample.

In addition to the SIEED, I use the BHP to analyze aggregate employment movements. The BHP is a 50% random sample of German establishments in the social security registry. It contains information on the number of employees by skill group, gender, part-time/full-time status, and age, as well as average compensation for full-time employees and selected moments of the within-establishment distribution (percentiles 25, 50, and 75). In line with the analysis of the linked employer-employee data described above, I restrict the sample to the time period 1999 to 2019.

County-Level Consumption Patterns: EVS Data by DESTATIS

To measure heterogeneity in consumption patterns, I rely on the *Income- and Expenditure Survey (EVS, short for Einkommens- und Verbrauchstudie)*, which is administered every 5 years by the Federal Statistical Office of Germany (DESTATIS). The EVS is a repeated cross-section that targets a representative sample of (depending on the survey wave) 42,000 and 55,000 households who are asked to keep detailed records of their income and expenditures for three months. Households are distributed equally across each quarter of the year. I use the EVS waves of 1998, 2003, 2008, 2013, and 2018. Expenditures are grouped into the 12 standard categories underlying the computation of consumer price indices: e.g., food and non-alcoholic beverages or shoes and clothing. Importantly,

[6] To provide a complete picture, Appendix A.4 presents estimates also for the period after 2019 alongside a discussion of what we can learn from the larger energy cost shocks in that period.

expenses for gasoline, gas, oil, and other energy goods are listed as separate items, allowing me to compute expenditure shares for commuting and different types of heating. Besides expenditures and income, the EVS also contains socio-demographic information on household members, such as occupational status, education, nationality, age, and some features of the living environment (apartment/house size in m^2, year in which the house of residence was built, rental vs. ownership status, ...). Finally, each household lists the available stock of durables such as bikes or cars.

Price Trends: Data from DESTATIS

While the EVS data allow me to measure differential exposure to energy price shocks across households, I rely on energy price series that are common to all consumers and represent the German average. DESTATIS provides a series of monthly average consumer prices for heating oil, gas, gasoline, as well as solid fuels, which I use as a price proxy for the residual *other energy* category in the EVS data, which includes coal, wood, other solid fuels, and central heating. I use yearly averages for each price series.

3 Empirical Approach

This section illustrates how the above-described data allows me to identify the pass-through of energy cost shocks into earnings. Section 3.1 introduces the empirical framework and discusses the role of the measurement of exposure to energy prices in identifying pass-through. Section 3.2 then provides descriptive evidence of exposure heterogeneity and the construction of the corresponding proxy for shocks to the cost of living.

3.1 Framework and Identification

Let $S^g_{i,t-s}$ be the share of energy good $g \in \{Gas, Gasoline, Oil, Other\}$ in individual i's total expenditures for the year $t-s$, and let P^g_t be the price of good g at time t. Following the classic Laspeyres approach, we can approximate the contribution of energy to changes in the cost of individual i's idiosyncratic consumption bundles by $C^s_{i,t} = \sum_g S^g_{i,t-s} \frac{P^g_t}{P^g_{t-s}}$. The goal of the empirical analysis is to relate this cost measure to changes in individual labor-market outcomes, particularly earnings.

As highlighted in the previous section, the employer-employee dataset underlying the analysis does not contain information on consumption. I, therefore, proxy $S^g_{i,t-s}$ with good g's expenditure share in individual i's county of residence, c. Consider the following regression,

treating the change in individual log earnings from year $t-s$ to year t as an outcome variable:

$$\Delta_s y_{i,c,t} = \alpha_i + \gamma_c + \delta_t + \tau \times C^s_{c,t} + \boldsymbol{X_{i,c,t}\beta} + \epsilon_{i,c,t}. \tag{1}$$

α_i, γ_c, and δ_t are individual, county, and year fixed effects, and $\boldsymbol{X_{i,c,t}}$ is a vector of individual$\times$year-specific covariates, discussed in further detail below.

Given the inclusion of time fixed effects and the definition of $C^s_{c,t}$ as the combination of price changes and expenditure shares, τ can be interpreted as the differential effect of energy price changes in counties with varying intensities of energy consumption. Using the change in log earnings as outcome, τ measures the share of direct additional real wage losses incurred by energy price increases that are compensated by changes in nominal earnings. For $\tau = 1$, county-specific losses are fully compensated; for $\tau = 0$, there is no correlation between additional costs and income changes. In the case of $\tau \in (0, 1)$, county-specific costs are partly recovered, but higher-expenditure counties suffer more substantial losses in real consumption than lower-expenditure counties. I will consider different time frames, s, to estimate short- and long-term responses.

Identification. To interpret τ causally as the effect of energy prices on wages through employees' consumption patterns, the energy price exposure measure $C^s_{c,t}$ needs to be (conditionally) uncorrelated with any unobserved variable that also affects wage changes for individual i in year t, i.e.,

$$\epsilon_{i,c,t} \perp\!\!\!\perp \sum_g S^g_{c,t-s} \frac{P^g_t}{P^g_{t-s}} | (\alpha_i, \gamma_c, \delta_t, \boldsymbol{X_{i,c,t}}). \tag{2}$$

The factor structure of $C^s_{c,t}$, i.e., the combination of local shares and global price shifters, renders this identification issue similar to the reasoning of shift-share instruments: the identifying variation for τ comes from the interaction of the two continuous measures and in turn any threat to identification needs to be correlated with this interaction term to be problematic. The recent work on shift-share instruments suggests either the exogeneity of shocks (here, $\frac{P^g_t}{P^g_{t-s}}$, Borusyak, Hull and Jaravel, 2022) or the exogeneity of shares ($S^g_{c,t-s}$, Goldsmith-Pinkham, Sorkin and Swift, 2020) as two main routes to identification.

A reason motivating the use of movements in energy prices as shocks to the costs of living and not prices for other goods is that energy prices are strongly influenced by trends on the global market and are thus plausibly determined at a higher level of aggregation than that of German

counties. One could consequently view $\frac{P_t^g}{P_{t-s}^g}$ as quasi-random from the point of view of a county. However, the absence of a causal relationship from local economic conditions to global prices does not necessarily imply that the two are uncorrelated. Without an explicit instrument for price changes, it is plausible that there exists a shock that is correlated with price trends and differentially impacts counties based on their energy expenditure shares. For example, since energy is mostly considered a necessity, its share of total expenditures decreases with income, making it more likely that higher-energy-expenditure counties are also poorer on average. If there exists any aggregate shock $Z_t \not\perp\!\!\!\perp \frac{P_t^g}{P_{t-s}^g}$ that differentially affects richer and poorer households, this would be an issue for the identification of τ.[7]

To strengthen the identification strategy, it is therefore helpful to also view the problem from the perspective of share exogeneity. This point of view implies that, even if unobserved factors drive both price and income changes, τ can still be recovered from the data as long as county-level energy expenditure shares are not correlated with other local conditions that moderate the effect of energy prices.

An example of a potential confounder would be a correlation between consumers' energy expenditures and the energy intensity of local production. In this case, the instrument could pick up both labor-demand and labor-supply responses. To account for such factors, $\mathbf{X_{ict}}$ contains a series of county- and individual-level characteristics and their interactions with δ_t. That is, the specification allows yearly shocks to vary over measures of employment concentration, county-level industry composition, unemployment rates, as well as indicators for the 3-digit industry of individual i's firm in period $t-s$, interacted with indicators for terciles of the within-sector firm size distribution. The latter implies that, in a given year, I compare earnings changes in response to energy price shocks for two individuals who, in $t-s$, worked for firms of similar size in the same sector but lived in counties with different exposure to the price shocks. This captures common annual shocks to firms with similar production processes, while allowing workers' exposure to energy price changes to vary.

To additionally account for the non-random sorting of households, I rely on adjusted county-specific expenditure shares rather than simple averages. As detailed in Section 3.2, I estimate energy expenditure shares for each county, $S_{c,t-s}^g$, adjusting for differences in household socio-

[7] One possibility to circumvent this issue is to instrument $\frac{P_t^g}{P_{t-s}^g}$ with policy surprises as in (Känzig, 2023). This approach relies on high-frequency price variation, however, which is less appealing in the setting of this paper, as wages are typically renegotiated at an annual or lower frequency. Section 4.1 nonetheless includes a specification that uses carbon price changes as an instrument.

demographic characteristics across space. The residual variation stems from geographic differences across Germany's 400 administrative counties, such as differences in commuting infrastructure or housing energy efficiency. The identifying assumption is that this residual geographic variation is not correlated with household characteristics that moderate the impact of price trends or correlated shocks.

3.2 Heterogeneous Exposure to Energy Price Shocks: Measurement and Descriptives

As just outlined, identification is partially based on variations in energy consumption patterns driven by geographic factors and not linked to socio-demographic characteristics such as income or education, which might themselves be important determinants of income changes. Using the EVS expenditure data, I therefore estimate for each survey wave a series of regressions of the type:

$$S^g_{h,c} = \pi_c + X_{h,c}\beta + u_{h,c}, \tag{3}$$

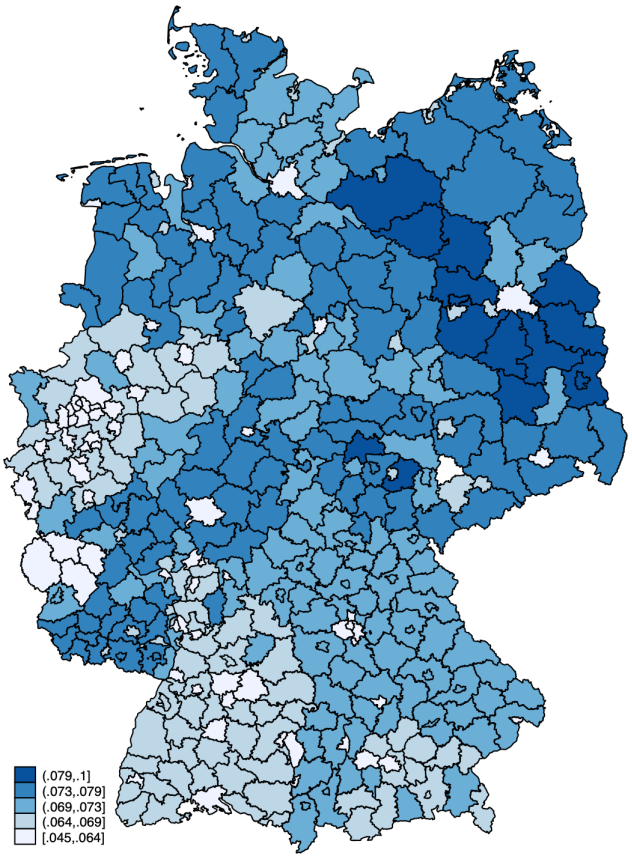
where $S^g_{h,c}$ is the share of expenditures that household h in location c devotes to good $g \in \{Gas, Gasoline, Oil, Other\}$, and π_c is a location c fixed effect. $X_{h,c}$ contains socio-demographics of the household, including the total number of household members, the number of employed household members, total expenditures, household gross income, living space per household member in m^2, the number of cars/household member, and the quarter of the year in which the expenditure documentation took place. Additionally, it includes the household head's marital status, occupation, full-time/part-time status, nationality, educational attainment, gender, and age.

The predictions obtained from (3) when fixing $X_{h,c}$ at its sample mean result in average expenditure shares for location c adjusted for socio-demographic differences between localities.[8] This, for instance, accounts for differences in energy expenditure shares that arise mechanically from the fact that energy is a necessity on which households in the upper tail of the expenditure distribution tend to spend a lower share. The remaining variation is based on locality fixed effects, which form the basis of my measure for exposure to energy price shocks.

To treat these estimated fixed effects as valid proxies in the spirit of the suggested empirical framework, I need to assume that, conditional on $X_{h,c}$, π_c does not contain information that predicts

[8] One complication in transferring the estimated location fixed effects to the IAB-FDZ data is that the EVS data contains geographic information at the region-by-agglomeration-type level and not directly at the level of an administrative county, which is the relevant level of aggregation in the IAB-FDZ data. I harmonize the two geographic measurements by relying on county-level characteristics obtained from the *Federal Institute for Research on Building, Urban Affairs, and Spatial Development (BBSR)*, as detailed in Appendix A.3.

Figure 1: Distribution of Energy Expenditure Shares across Counties



Note: The figure shows the spatial distribution of average (across all five EVS waves) estimated county-level total expenditure shares of energy goods, including gasoline, gas, oil, and other types of energy. For an overview of the distribution across all years, see Figures A1 and A2. Source: Einkommens- und Verbrauchsstudie 1998, 2003, 2008, 2013, and 2018, each Basic File 3, own calculations.

both the response of income to energy cost shocks (e.g., through differences in bargaining power) and a higher energy expenditure share. Importantly, note that $X_{h,c}$ contains household income. This implies that even if unobserved differences between households across space exist and are absorbed into π_c, this would only be problematic if they are predictive of income changes in response to energy prices but not of income levels.

Descriptive Evidence for Heterogeneity in Energy Price Exposure

Predicting county-level expenditures using specification (3) and EVS data reveals significant spatial heterogeneity, even when controlling for household socio-demographic characteristics. Figure 1 displays the distribution of energy expenditure shares, defined as the sum of expenditure shares over all types of energy g. While energy expenditures are only 4.6% of total expenditures in some

Table 2: Expenditure Shares for Different Energy Types (in %)

	Mean	*S.D.*	*Min*	*Max*
Gasoline	3.88	0.60	2.38	6.64
Gas	1.31	0.50	0.31	2.75
Oil	0.72	0.39	0.03	1.97
Other Energy	0.99	0.38	0.03	2.00
All Energy	6.91	1.08	4.57	9.78

Notes: The table shows estimated county-level averages for expenditure shares of different types of energy goods. *Other Energy* includes expenditures for coal, wood, and other solid fuels, as well as central heating. Source: Einkommens- und Verbrauchsstudie 1998, 2003, 2008, 2013, and 2018, each Basic File 3, own calculations.

counties, they amount to almost 10% in others (see also Table 2). This difference is comparable to the gap in expenditure shares between the highest- and lowest-income household quintiles. On average, 6.9% of expenditures are allocated to energy consumption, with a standard deviation across counties of around 1.1 percentage points.

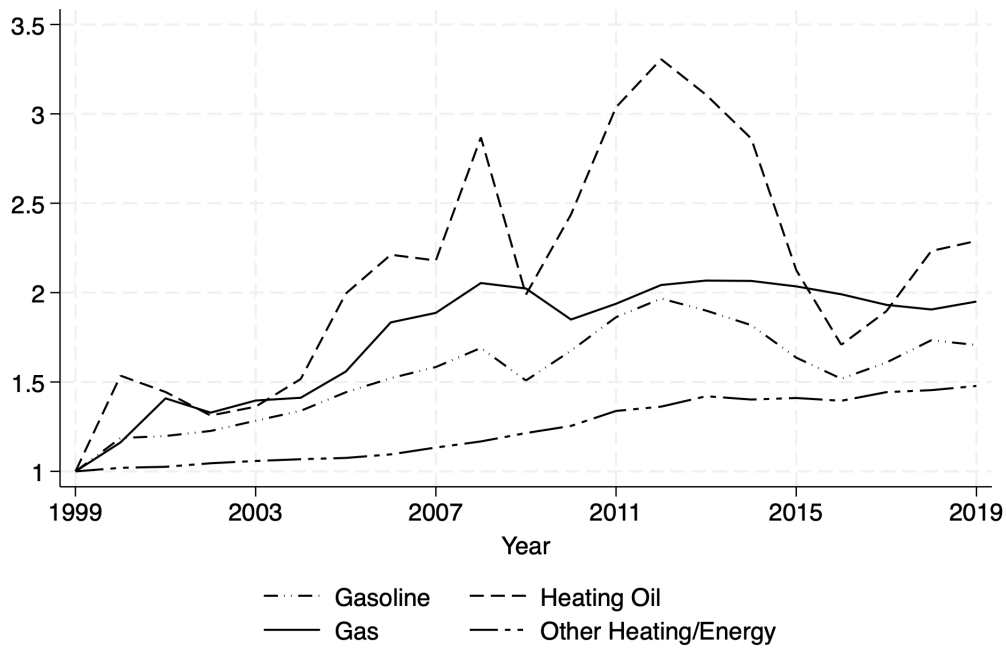
Also within finer categories of energy consumption, there is considerable variability across counties. Table 2 breaks energy expenditures down into expenditures on gasoline, gas, oil, and other types of energy. The most significant component is gasoline consumption, which, on average, accounts for 3.9% of total expenditures, whereas oil is the lowest-expenditure category, with an average share of 0.7%. However, even in this category, some counties have average households that spend 2% of their expenditure budget on oil. Judged by the ratio of standard deviation to mean, oil expenditures are the most variable form of energy consumption across space, followed by gas and the residual category of other energy types, consisting of coal, wood, and other solid fuels, as well as central/district heating.

By construction, this spatial variation in expenditure shares is not driven by differences in household sorting based on average income, age, education, or labor force status. Using information on county-level characteristics provided by the Federal Institute for Research on Building, Urban Affairs, and Spatial Development (BBSR), however, reveals that energy expenditure shares are higher in counties with lower population density, a larger share of commuters, a sparser public transportation network, and a lower frequency of construction of new housing (see Table A1). All these features point in the direction that the variation in energy expenditure shares captures significant spatial heterogeneity in commuting and housing infrastructure. Where the housing stock is older and people need to use cars more for commuting, energy expenditure shares are

higher.

This demonstrates a significant degree of heterogeneity in consumers' exposure to changing energy prices. Figure 2, on the other hand, displays the magnitude of these changes by plotting the time series of average German consumer prices for different energy goods over the sample period. Relative to 1999, prices for gas and heating oil doubled by 2019, and also prices for gasoline and other energy types increased considerably more than expected from average inflation. Overall CPI inflation for the same period is approximately 34%. Figure 2 further illustrates a non-negligible amount of variation around the overall positive time trend, in particular for gasoline, oil, and gas. This implies that the identification of cost-wage pass-through will not rely solely on monotonically increasing prices.

Figure 2: Consumer Price Series for Energy Goods



Note: The figure plots a series of average yearly consumer prices in Germany, normalized to 100 in 1999. Source: DESTATIS.

One motivation for using energy prices as shocks to the cost of living rather than price increases for other goods is that energy prices are plausibly determined at a higher level of aggregation than German counties. This both facilitates the exclusion of unobserved factors that jointly determine local economic trends and cost increases, and it implies that a common energy price series is a reasonable approximation of local costs. Unfortunately, there is limited county-level

price data to corroborate this latter assumption.[9] The latest available information at this level of granularity stems from the period 2007 to 2009 (approximately the midpoint of the period under study in this paper) and was collected as part of a project by the BBSR (2009). While the BBSR study demonstrates remarkable differences in local price levels (assuming a common consumption basket but using varying prices), it also shows that the variation in prices for gasoline and heating oil is minuscule, with cross-county coefficients of variation below 5%. This is less than a quarter of the variation in the price of local public transport, for instance, and one-tenth of the variation in the cost of placing advertisements in local newspapers (see Figure A3 for an overview). I conclude that the assumption of a common energy price series is reasonable.

4 Empirical Results

The empirical results of this paper are presented in three steps. Combining the above-described measure of county-specific cost shocks with employer-employee data, I first demonstrate that nominal earnings respond positively to energy price shocks. In Section 4.2, I then discuss the role of job-to-job mobility in mitigating real consumption losses. Section 4.3 considers other on-the-job adjustment margins as well as the extensive margin of non-employment.

4.1 Energy Price Changes and Average Earnings Responses

This section present estimates of specification (1), using county-specific energy-driven cost increases, $C^s_{ct} = \sum_g S^g_{c,t-s} \frac{P^g_t}{P^g_{t-s}}$, as a proxy for individual heterogeneity in shock intensity. The regressions include individual and county fixed effects, such that the model identifies τ from deviations from idiosyncratic time trends in earnings and additionally adjusts for year-by-sector by within-sector-firm-size tercile fixed effects. Standard errors are clustered at the county level.

Table 3 presents estimates for year-to-year ($s = 1$) changes in nominal earnings following changes in the local energy-specific CPIs. The results indicate a non-negligible degree of pass-through. Based on the results in column (1), a one standard deviation higher energy cost shock is associated with a 0.4 percentage point higher growth in nominal earnings. For comparison, the yearly mean nominal earnings growth over the considered sample period is 3%. An average county that dedicates 6.9% of its total expenditures to energy experiences a 0.69% increase in costs when

[9] Of course, as discussed above, the assumption of common price series across space is not strictly needed as long as the variation in energy expenditure shares is uncorrelated with other local mediators of global shocks.

Table 3: The Effect of Year-to-Year Energy Cost Shocks on Earnings

	$Outcome = ln(Earnings_{it}/Earnings_{i,t-1})$				
	(1)	(2)	(3)	(4)	(5)
Energy Cost Shock	0.334***	0.333***	0.286***	0.358***	0.303***
	(0.050)	(0.049)	(0.046)	(0.057)	(0.046)
N Total	7,101,140	7,101,140	7,101,140	7,101,140	7,101,140
N Individuals	639,016	639,016	639,016	639,016	639,016
Match HHI		✓			✓
Match Energy-Intensity			✓		✓
Match Unemployment				✓	✓

Notes: The Table displays estimates for τ in specification 1, proxying individual energy-price inflation with county-level averages as estimated in Section 3.2: $C^1_{ct} = \sum_g S^g_{c,t-1} \frac{P^g_t}{P^g_{t-1}}$. Each specification includes individual and county fixed effects, as well as year-by-sector-by-within-sector-firm-size-tercile fixed effects. Firm size is measured by the number of employees in the firm that individual i worked for in year $t-1$. Columns (2) to (4) allow yearly fixed effects to vary over: quintiles of the county-sector-level distribution of employment concentration (Herfindahl-Hirschmann-Index), quintiles of the energy intensity of local production, measured as the share of establishments in each county that are operating in energy-intense sectors (i.e., agriculture, utilities, transportation, manufacturing), and the lagged county-level unemployment rate, respectively. Standard errors are clustered at the county level.

energy prices rise by 10%. Another way to think about the magnitude of the estimated coefficient is that 33% of the additional real losses in purchasing power can be recovered within the same year through higher earnings, i.e., earnings increase by 0.23%. While this still implies a divergence in real earnings between high- and low-energy-expenditure areas, the gap is narrowed by endogenous labor-market adjustments.

Columns (2) through (5) of Table 3 show that the results are comparable across different specifications that allow for common yearly shocks across labor markets with similar characteristics. These include the county-sector-level concentration of employment (Herfindahl-Hirschmann Index, column 2), the county-level share of establishments operating in energy-intensive sectors (agriculture, utilities, transportation, manufacturing; column 3), and the lagged county-level employment rate (column 4). This stability lends credibility to the identifying assumption that county-level expenditure shares are uncorrelated with other local mediators of global shocks.

Appendix Tables A2 and A3 contain a large number of additional robustness checks. The pass-through estimates are robust to instrumenting yearly energy cost shocks with carbon policy surprises taken from Känzig (2023) and to using changes in top-coded rather than imputed earnings as the outcome variable. The results are also unaffected when fixing expenditure shares, S^g_c, at their

earliest available level (i.e., 1998) rather than updating shares dynamically over time, and when constructing the energy cost instrument using a broader definition of local labor markets rather than relying on administrative counties. For the latter set of specifications, I define local labor markets in accordance with the classification of the Federal Institute for Research on Building, Urban Affairs, and Spatial Development (BBSR), which takes common commuter flows into account and reduces the number of local labor markets to 223 (see Table A3). As further robustness checks, I show that the results are not affected by the inclusion of macro-region (*Bundesländer*) by year fixed effects. That is, the energy cost instrument is not spatially clustered in a way that captures differences in macro-level time trends across German regions (see Table A2).

The Role of Labor Demand. An identifying assumption of the empirical analysis posits that, conditional on all included covariates, local energy expenditure shares are not correlated with other mediators of global energy price movements. While the discussion above already addresses several potential confounders, a clear mediator that would confound the estimation is responses in local labor demand. If higher energy prices have a disproportionately negative impact on the labor demand of energy-intensive firms, and more energy-intensive firms are located where consumers spend a larger share of their budgets on energy, then the previously documented pass-through would be a lower bound. The opposite would hold if consumers' energy expenditure shares and firms' energy intensity were negatively correlated.

To address this issue, columns (3) and (5) of Table 3 adjust for yearly shocks to counties with a similar industry composition. All specifications additionally compare only individuals who worked for firms of similar size in the same 3-digit sector in the previous year. The identifying assumption is that the remaining variation within the 3-digit sector and firm-size cells is uncorrelated with consumers' energy expenditure shares.

While the SIEED data does not contain information to test this assumption directly, there are reasons to believe that potential violations of it are likely to be of limited quantitative relevance. For instance, von Graevenitz and Rottner (2023) estimate that the share of energy in the total production costs of German manufacturing firms is low, averaging around 2-3% for the period 2003 to 2017. Most of it is driven by gas purchases. Reassuringly, Table A4 in the Appendix demonstrates that excluding gas from the energy cost shock measure does not affect the overall pass-through estimates.[10] Petrick, Rehdanz and Wagner (2011) further provide estimates for the set of 3-digit

[10]Incidentally, this result also confirms that no single energy subcategory is driving the results, which is a common

sectors with the highest within-sector variation in energy intensity.[11] This is the relevant metric if one is concerned about a correlation with spatial differences in consumers' energy expenditures, since all regressions above already compare only individuals who work in the same 3-digit sector. In Appendix Table A5, I show that excluding the counties with the highest and lowest share of employment in these sectors does not affect the estimated pass-through.

As the final and most restrictive test of the relevance of an alternative labor demand channel, I estimate a model that compares only workers who worked in the exact same establishment in period $t-1$. The remaining variation in energy expenditure shares across workers arises because some workers commute outside their county of residence for work, and the energy cost instrument is assigned at their place of residence. While this specification excludes many workers who are employed in establishments without (out-of-county) commuters, the results remain qualitatively similar, indicating a substantial pass-through of energy cost shocks into nominal earnings ($\hat{\tau} = 0.27, p < 0.01$, see Table A2). The accumulation of evidence presented in this section speaks against a quantitatively relevant bias arising from unobserved labor-demand responses.

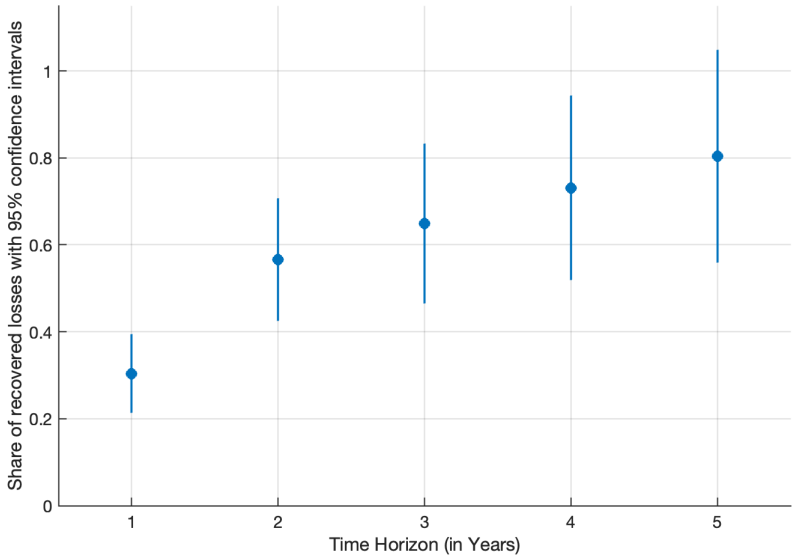
Dynamic Adjustments: Considering cumulative exposure over longer-time horizons. It is natural to assume that labor market adjustments occur dynamically and not necessarily on a short-term basis – e.g., due to mobility frictions or a lower frequency of wage adjustments. To understand the pass-through of energy prices to earnings on a longer-term horizon, Figure 3 displays the point estimates obtained by repeating the analysis for cumulative changes over periods of one to five years. Considering longer time horizons reveals an increasing pass-through over time, such that over five years, earnings increases recover around 80% of county-specific cost shocks for the average employee (see Table A6 for point estimates and standard errors). The largest change in pass-through, and the only year-on-year difference in point estimates that is statistically significant, happens when moving from a one-year to a two-year comparison.

Over a longer time horizon, labor market dynamics substantially reduce the loss in real consumption that high-exposure areas would have experienced relative to low-exposure areas in a counterfactual in which nominal earnings moved in proportion to average energy cost changes.

sanity check for instruments that rely on a shift-share logic for identification (Goldsmith-Pinkham, Sorkin and Swift, 2020).

[11] The list of sectors is (i) mining, (ii) glass, (iii) coke, petroleum products, and nuclear fuel, (iv) locomotives and rolling stock, (v) non-metal recycling, (vi) basic and agro-chemicals, (vii) metal recycling, (viii) basic iron and steel, tubes, (ix) veneer sheets, plywood, laminboard, and the like, and (x) builders' carpentry and joinery.

Figure 3: The Effect of Energy Cost Shocks on Earnings; for varying time-horizons



Note: The Figure displays point estimates and 95% confidence intervals for τ in specification 1, for varying time-horizons $s \in \{1, 2, 3, 4, 5\}$. Each specification includes individual and county fixed effects, as well as year-by-sector-by-within-sector-firm-size-tercile fixed effects. Firm size is measured by the number of employees in the firm that individual i worked for in year $t - s$. The specifications additionally allow yearly fixed effects to vary over: quintiles of the county-sector-level distribution of employment concentration (Herfindahl-Hirschmann-Index), quintiles of the energy intensity of local production, measured as the share of establishments in each county that are operating in energy-intensive sectors (i.e., agriculture, utilities, transportation, manufacturing), and the lagged county-level unemployment rate, respectively. Standard errors are clustered at the county level.

Note that, in the spirit of focusing on labor market dynamics, the key variable of interest is a Laspeyres price index, which fixes lagged consumption patterns. Naturally, a second margin of adjustment is a change in consumption behavior, which can provide further buffer against increasing costs. Auer et al. (2024) explicitly discuss this in the context of changing relative prices between imported and domestic goods in Switzerland, and Ahlvik et al. (2025) discuss Finnish household consumption responses to the European energy crisis of 2022.[12]

Transitory vs. Persistent Price Changes. One question when thinking about workers' responses to changes in energy prices is whether they consider price changes to be persistent. Clearly, if price changes were fully transitory, then taking costly actions on the labor market would not be an appealing strategy for most employees. While the SIEED data does not include surveys that would

[12] A second margin to consider when analyzing longer time horizons is the pass-through of energy prices into the prices of other consumption goods. Hintermann et al. (2024) estimate that German manufacturing firms pass-through around 30% of energy costs, which based on estimates by von Graevenitz and Rottner (2023) constitute between 2 and 3% of production costs in the German manufacturing sector. The scope for other prices to significantly affect the conclusions drawn above is thus limited.

allow me to measure workers' expectations, the empirical analysis purposefully focuses on year-to-year changes in energy prices. That is, the paper abstracts from high-frequency price variation over days, weeks, or months, where changes are more often transitory and, in fact, documents substantial labor market responses. The above-documented increase in pass-through over longer time horizons further supports the idea that, over time, employees likely consider price changes to be persistent.

A related debate is whether price changes are expected or unexpected from the employees' perspective. Since price changes are serially correlated, employees might correctly expect higher prices today to be associated with larger future price changes. While this is not a threat to identification, which, as discussed, also relies on the quasi-random assignment of expenditure shares across counties, it would change the interpretation of the estimated pass-through to one that measures not only labor market responses to realized inflation but also to expected inflation. Pilossoph and Ryngaert (2025) show, using U.S. survey data, that consumers with higher inflation expectations engage more intensely in job search. In fact, when instrumenting energy price changes with carbon policy surprises, the estimated pass-through increases. This suggests that for expected changes, part of the pass-through is likely captured by a lagged version of the instrument, providing a further rationale for why pass-through estimates increase with longer time horizons.

Treatment Effect Heterogeneity. This paper mainly discusses the average difference in the response to energy price changes across local labor markets with varying energy expenditures. The reason is that, for identification, I proxy energy cost shocks with county-level CPIs that are adjusted for socio-demographic variables. When looking at differences across subgroups, it is therefore not straightforward to cleanly separate treatment-effect heterogeneity from heterogeneity in the cost shock/the treatment. With that caveat in mind, a subgroup analysis nonetheless reveals some noteworthy patterns, discussed in detail in Appendix A.4.

For example, younger workers are considerably more likely than their older peers to experience increases in nominal earnings after energy cost shocks. Employees aged 25 to 40 can recover 46% of county-specific cost shocks within a year, whereas the corresponding figure for the age group 41 to 55 is 13%. At the same time, they are considerably more likely to switch jobs, suggesting that job-to-job mobility can be a route to recovering losses. While this channel is discussed in detail in the next section, the heterogeneity analysis also cautions that it is not the main channel for all workers. For instance, employees with a college degree recover a somewhat

higher fraction of predicted losses in real consumption than those without a college degree (35% vs. 28%), even though the latter group has a stronger job-mobility response to energy-cost shocks.

4.2 The Role of Job-to-Job Mobility in Mitigating Earnings Losses

A series of recent studies suggests that switching employers might be one channel through which employees can recover real earnings losses after price changes (Afrouzi et al., 2026; Bostanci, Koru and Villalvazo, 2025; Pilossoph and Ryngaert, 2025). In line with this idea, Column (1) of Table 4 presents estimates of the effect of energy cost shocks on the probability of employer-to-employer (E-E) transitions.

Employees in counties with a 1% higher energy cost shock experience a 0.3 percentage-point increase in the probability of switching employers. To put this into context, on average, 16% of the estimation sample switches jobs in a given year. While non-negligible, the estimated elasticity is thus moderate relative to the typical labor mobility rates in Germany. However, Table 4 further highlights that average job switches are a poor description of job transitions in high-exposure counties. Columns (2) and (3) present estimates of the link between energy cost shocks and earnings for the sub-samples of individuals who, respectively, did and did not switch jobs between period $t-1$ and t. Conditional on switching jobs, employees' earnings gains offset an additional half of energy-driven cost shocks on a year-to-year basis. Note that the comparison in column (2) is between two individuals who switch their jobs between years $t-1$ and t but differ in their exposure to energy price increases. While this conditioning on a post-treatment variable implies that the estimated effect is not the causal effect of energy cost shocks on changes in nominal earnings among switchers, the estimate is informative of the selection at play. Energy cost shocks not only encourage employees to switch jobs, which, on average, entails a significant increase in earnings, but also make job switches more closely correlated with earnings gains. That is, job switches in high-exposure counties are positively selected.

A second key takeaway from Table 4 is that a positive pass-through of energy prices into earnings is not constrained to the subsample of job switchers alone. Also those employees who stay in their current jobs experience a positive earnings response after increases in energy costs, albeit to a more minor degree (column 3). One way to think about the estimated magnitude is that it is consistent with around 1 in 4 employees in high-exposure counties fully recovering from cost shocks through higher earnings despite remaining on the job. Using the average difference in yearly earnings changes between job switchers and stayers (8.2%-2.4%=5.8%), together with the

Table 4: The Effect of Energy Cost Shocks on Job-to-Job Mobility

	Pr(E-to-E)	$\Delta ln(earnings)$	
	(1)	(2)	(3)
Yearly Energy Cost Shocks	0.273*	0.522***	0.277***
	(0.161)	(0.125)	(0.039)
Sample	Full	Switchers	Stayers
N Individuals	639,034	442,308	634,913
N Total	7,115,473	1,150,726	5,964,747

Notes: The Table displays estimates for τ in specification 1, proxying individual energy-price inflation with county-level averages as estimated in Section 3.2: $C^s_{ct} = \sum_g S^g_{c,t-s} \frac{P^g_t}{P^g_{t-s}}$. Column (1) uses an indicator that is equal to 1 in case individual i switched their employer between $t-s$ and t as the outcome variable. Columns (2) and (3) use changes in log earnings as outcomes and restrict the estimation to the sample of job switchers and stayers, respectively. Included covariates are identical to Table 3. Standard errors are clustered at the county level.

estimated increase in the selectivity of job switches, a naive back-of-the-envelope decomposition suggests that job mobility explains around one-third of the documented average earnings response. The remainder is due to changes on the job.

Characterizing Job Mobility after Energy Cost Shocks. Section 4.3 below presents a discussion of why job stayers might be compensated for higher energy costs. Before that, the remainder of this section, however, presents evidence on the nature of job switches after energy cost shocks. Table 4 highlights that on a year-to-year basis, selected E-E transitions are an important buffer against energy cost shocks. This is not trivial, given that cost shocks make transitions more "local". A one-standard-deviation increase in energy costs reduces the probability that employees commute out of their county of residence to work by 0.5%. Previous research by Agrawal, Jahn and Janeba (2024) demonstrates that when job seekers' geographic job-search radius increases (in their case, in response to more generous commuting subsidies), wages do increase as well. The fact that I find both a reduction in commuting and an increase in earnings suggests that average job switches differ for individuals who have been harder hit.[13]

To get a better understanding of why that is, Table 5 uses the subsample of employees who changed their jobs between year $t-1$ and t to estimate models similar to specification (1), treating characteristics of the newly joined firms as the outcome variables. Column (1) indicates that job

[13]The finding that commuting distances decrease in response to higher energy costs is in line with Ready, Roussanov and Zurowska (2019) who study commuting patterns after increases in oil prices for a sample of employees in Minnesota. Besides its implications for labor market dynamics, this finding also suggests that employees respond along the consumption margin. Reducing commuting likely reduces gasoline expenditures, providing an additional buffer against rising living costs.

Table 5: Energy Cost Shocks and the Characteristics of Switchers' New Firms

	Exp. Change in:	Probability of:	Pr(Switch to Different:)		
	$ln(earnings)$	$\Delta ln(earnings) < 0$	Sector	Occupation	Task
Energy Cost Shock	0.755***	-0.648***	0.913***	0.829***	-0.0212
	(0.129)	(0.208)	(0.139)	(0.192)	(0.134)
N Individuals	382,780	442,052	442,308	442,308	436,611
N Total	850,815	1,148,421	1,150,727	1,150,727	1,099,468

Notes: The Table displays estimates for τ in specification 1, proxying individual energy-price inflation with county-level averages as estimated in Section 3.2: $\sum_g S^g_{c,t-1} \frac{P^g_t}{P^g_{t-1}}$. Included covariates are identical to Table 3. Expected change in $ln(earnings)$ is defined as the difference in log average earnings between individual i's current employer and their employer in $t-1$, measured in the year $t-1$. The probability of switching to a different task is equal to 1 if the current job task of individual i is more complex than the performed task in $t-1$. Job complexity is measured at the occupational level and can take four values: unskilled/semiskilled, skilled, complex task, and highly complex task. The probability of switching to a different sector or occupation is based on 3-digit classifications. Included covariates are identical to Table 3. Standard errors are clustered at the county level.

switchers in higher-exposure counties are more likely to sort into firms with higher overall earnings potential. The outcome variable in this specification is the difference in log earnings between the average employee at individual i's current firm and the average employee at their previous firm, based on information from year $t-1$. This way, the resulting log difference provides an estimate of the change in earnings an average person can expect simply by joining their new firm, irrespective of bargaining efforts during the switch. Interestingly, the estimated differences are even larger than the overall pass-through among switchers. While workers join better-paying firms, their rank in the within-firm pay distribution at the new firm is thus lower than at their previous firm.

In line with the idea that workers in high-exposure labor markets are more likely to switch to higher-paying firms, column (2) of Table 5 shows that the likelihood of observing an E-E transition associated with lower earnings drops substantially. In my sample, around one-third of all job transitions are associated with a decrease in earnings. The joint results in columns (1) and (2) are informative for thinking about how to structure an economic theory to rationalize the observed patterns. While a model of endogenous search would be sufficient to generate increased selectivity of job switches, voluntary transitions in such frameworks are typically associated with weakly increasing wages. If, on the other hand, workers also care about non-wage firm attributes, voluntary negative earnings transitions can easily be rationalized (e.g., Becker, 2011; Hall and Mueller, 2018; Sullivan and To, 2014).[14] Sorkin (2018) uses this notion precisely to infer the valuation of a bundle

[14]An alternative reason for voluntary transitions that are associated with negative earnings changes are transitions

of such firm attributes from workers' revealed preferences. In Section 5 below, I provide a simple theoretical framework following up on that idea. In it, workers with greater exposure to energy cost shocks align their employer choice more closely with earnings considerations and place less weight on non-pecuniary factors. This makes their transitions positively selected for earnings gains.

Columns (3) and (4) of Table 5 show that job switchers in high-exposure counties also display a higher willingness to join firms in different sectors and to accept jobs in different occupations. Column (5) finally demonstrates that while workers switch occupations and sectors, their earnings increases are unlikely to be explained by a transition to more complex jobs. The SIEED data provides a measure of task requirements on the job based on job tasks associated with different occupations. Employees in high-exposure counties are not more likely to change to a job with a higher skill requirement than in their previous employment spell.

The Dynamics of Job Switching. Section 4.1 demonstrates that the pass-through of energy cost shocks into nominal earnings increases over time. Given the evidence presented that selected job mobility plays an important role in short-run earnings gains, a natural question is how much of the increasing pass-through can be accounted for by growing flexibility of workers towards E-E transitions. On a year-to-year basis, a 1% energy-driven increase in costs leads to a 0.3 percentage-point higher probability of job-to-job mobility. The same holds over a five-year window (see Table A7 for point estimates and standard errors).

At the same time, over a five-year period, a larger share of employees switch jobs regardless of energy costs. While the average yearly job mobility rate is 16%, 43% of all employees switch jobs over 5-year windows. Against this larger baseline, the response of job mobility to energy prices is comparatively small. However, also over longer time horizons, job switching is associated with considerably higher earnings gains in high-exposure labor markets (for a five-year window, $\hat{\tau} = 1.26, p < 0.01$, see Table A7). In combination with the overall higher job mobility over longer time periods, this maintained selectivity pushes the average pass-through upwards.

4.3 Other Margins of Adjustment to Cost Shocks

The previous section demonstrates that selected job-to-job mobility is an important mechanism underlying the partial recovery of real earnings among employees. But, the results of Table 4

due to intertemporal trade-offs of short-term earnings losses with dynamic gains (Postel-Vinay and Robin, 2002). While intriguing, the theoretical model presented below abstracts from such dynamics. It focuses on medium-term adjustments over 5-year windows, where such strategic short-term losses are plausibly of lower quantitative relevance.

also indicate adjustments in nominal earnings among job stayers. One way in which both might be linked is if workers generate more outside offers after energy cost shocks and use them as leverage to re-bargain wages on the job. Pilossoph, Ryngaert and Wedewer (2023) explore this option theoretically in the context of general inflation, and Caldwell, Haegele and Heining (2024) provide evidence that individual-level bargaining has become more prevalent in Germany.

Since the ability to generate outside offers is plausibly greater among workers with a college degree, the fact that pass-through is stronger among job stayers with a college degree largely supports the idea that bargaining is one potential mechanism. Note that such bargaining need not occur at the individual level. As discussed by Jäger, Noy and Schoefer (2022), worker councils are an integral part of the German system of co-determination and often engage in firm-level bargaining in addition to union-led sector-level agreements.

Table 6 presents the results of specifications that test other plausible channels besides job mobility that could explain a positive earnings response. For instance, the outcome variable underlying the documented nominal earnings response is total daily earnings and not earnings per hour worked. An alternative margin of adjustment is consequently an increase in working hours. Unfortunately, the SIEED data does not include detailed hours measurements beyond an indicator for part-time status. However, the main analysis sample in this paper consists only of full-time workers, for whom variation in hours worked is relatively limited in Germany, and there is no evidence that overtime pay plays a quantitatively relevant role. Workers are no more likely to receive a bonus payment (Column 2), which IAB-FDZ defines as any non-standard amount paid in addition to regular salary, including overtime pay. More generally, Zapf and Weber (2017) estimate that only around 6.1% of German full-time workers receive any overtime pay at all. Sectors in which overtime is more frequent are agriculture, wholesale, retail, transportation, and accommodation services. In the manufacturing and construction sector, on the other hand, paid overtime is less likely. I do not find a higher pass-through of energy cost shocks into earnings in the former set of sectors than in the latter, even when allowing yearly fixed effects to vary over a host of worker characteristics that are predictive of worker sorting into both sets of sectors (Table 6, columns 3 and 4).

Table 6, Column (2) further shows that workers do not "up-skill". That is, I find no evidence of promotions that could explain the rise in nominal earnings and that might have resulted from increased effort after cost increases. I interpret the absence of results favoring hours adjustments or effort-induced bonus payments as suggestive evidence that bargaining is a key mechanism for

Table 6: Energy Cost Shocks and the Characteristics of Stayers

	Outcome =1, if individual i:			
	Received a Bonus	Switched Task	Outcome: $\Delta ln(Earnings)$	
	(1)	(2)	(3)	(4)
Energy Cost Shock	0.050	0.101	0.366***	0.343***
	(0.046)	(0.084)	(0.067)	(0.069)
N Individuals	639,076	636,544	265,467	219,980
N Total	7,116,270	6,909,404	2,425,527	1,803,701
Sample	Full	Full	OT Uncommon	OT Common

Notes: The Table displays estimates for τ in specification 1, proxying individual energy-price inflation with county-level averages as estimated in Section 3.2: $\sum_g S^g_{c,t-1} \frac{P^g_t}{P^g_{t-1}}$. The probability of switching to a different task is equal to 1 if the current job task of individual i is more complex than the performed task in $t-1$. Job complexity is measured at the occupational level and can take four values: unskilled/semiskilled, skilled, complex task, and highly complex task. Covariates for columns (1) and (2) are identical to Table 3, column (5). Columns (3) and (4) are estimated separately for individuals who in $t-1$ worked in sectors in which overtime is relatively uncommon (manufacturing and construction) or relatively common (agriculture, wholesale, retail, transportation, and accommodation services), respectively. Each specification includes individual and county fixed effects, as well as year-by-sector by within-sector firm-size tercile fixed effects. The models additionally allow yearly fixed effects to vary with an individual's propensity score for working in sectors where overtime is relatively common. The propensity score is estimated using a probit model with individual age, gender, education (university degree vs. no university degree), county of residence, citizenship (German vs. non-German), and the task complexity of the current job as predictors. Standard errors are clustered at the county level and displayed in parentheses.

earnings adjustments in addition to targeted job mobility. In a survey of U.S. workers, Guerreiro et al. (2024) find evidence of increased willingness to negotiate with employers when inflation is high. While there is no explicit bargaining in the theoretical model I present below, firms take workers' outside options and the credibility of their willingness to switch jobs into consideration when posting wages. This implicitly captures the idea that the distribution of outside offers shapes the pass-through a worker can expect when staying on the job.

The Extensive Margin of Labor Supply. So far, the empirical analysis has neglected changes in the extensive margin of employment. One way to rationalize a positive response of nominal earnings to increases in energy prices is a contemporaneous decline in labor force participation, which raises the marginal productivity of labor. If this occurs to a greater extent in high-exposure labor markets, my estimates might capture extensive-margin changes rather than the actions of workers who remain employed.

To investigate this channel, I examine changes in aggregate employment using BHP firm-

Table 7: Effect on County-Level Employment

	$\Delta ln(\#employees)$	UE Rate
	(1)	(2)
Energy Cost Shock	0.216**	-0.080***
	(0.102)	(0.016)
N Counties	400	400
N Total	8,000	8,000
Data	IAB-BHP	BBSR

Notes: Energy cost shocks are defined as county-level energy-specific CPIs, as estimated in Section 3.2: $\sum_g S^g_{c,t-1} \frac{P^g_t}{P^g_{t-1}}$ The outcome in column (1) is the change in log-employment relative to the previous year, whereas the outcome in column (2) is the unemployment rate in county c. The regression adjusts for county fixed effects and year-by-quintiles of *Energy Intensity* fixed effects, where energy intensity is defined as the share of establishments operating in energy-intensive sectors (i.e., agriculture, utilities, transportation, manufacturing). Standard errors are clustered at the county level.

level data and BBSR unemployment data. Since results using log-employment as the outcome are highly sensitive to changes in employment among small firms with one or two employees, I aggregate firm-level employment to the county-year level. In line with the analysis of employer-employee data presented earlier, I regress changes in log employment in county c and year t relative to year $t-1$ on the county-specific energy CPI and county fixed effects. The regressions also allow yearly fixed effects to vary across quintiles of production energy intensity. As before, standard errors are clustered at the county level.

Table 7 contains the results. Contrary to what would be needed to rationalize an increase in earnings based on the marginal productivity of labor, aggregate employment responds weakly positively to energy cost shocks. A 1% increase in costs leads to a 0.2 percentage point increase in county-level employment as represented by the BHP. Note that the BHP does not cover self-employment. The inflow can consequently come out of self-employment, unemployment, or non-participation in the labor market. Column (2) shows that county-level unemployment correspondingly drops in high-expenditure areas relative to low-expenditure areas.

Bloesch, Lee and Weber (2024) argue that a quit into unemployment is not a credible threat to nudge firms towards increasing nominal earnings after cost shocks. In line with this, the results in this section show that, if anything, there is a flow out of unemployment when energy costs increase. Thinking about this in terms of a standard labor-supply model again suggests that the income effect dominates the substitution effect, and that workers are more willing to bear the disutility of labor when their real earnings decrease. In the next section, I present a theoretical framework to provide some intuition for the empirical results.

5 A Theoretical Assessment

This section discusses a theoretical framework linking exogenous movements in energy prices to individuals' labor-market outcomes. The fact that we observe worker responses to energy cost shocks that lead to higher nominal earnings implies that these adjustments are not costless. However, in models in which workers trade off the utility derived from wages (or consumption) with that of non-pecuniary job aspects, we can think about the empirically documented shocks to the cost of living as triggering income effects that make workers more willing to pay for higher earnings in the form of the non-pecuniary utility component. Such non-pecuniary costs could include, for instance, the disutility of working longer hours, search costs, or the non-pecuniary amenities associated with a given job.

Motivated by an empirical analysis that leaves little role for intensive-margin labor supply decisions and focusing on longer-term adjustments that abstract from short-term search frictions, this paper follows the latter approach and considers a simple equilibrium model of the labor market in the style of Card et al. (2018). Workers have heterogeneous preferences over firms' pecuniary and non-pecuniary attributes, and firms post wages. Into this setting, I embed a consumption problem with non-homothetic preferences over energy and a residual consumption good. Below, I first outline the model and its mechanics, and then estimate its key parameters by targeting the empirical differences in earnings adjustments and job-to-job transition rates over five-year windows. Importantly, the framework explicitly models a labor market equilibrium but is a partial equilibrium with respect to the product market.

5.1 Laying Out the Model

Consider a set of local labor markets $l = 1, 2, ..., L$, each inhabited by a mass of workers, N_l, and a finite number of firms $j \in J_l$. The location of workers and firms is fixed, and workers cannot work for firms outside their local labor market. Their outside option is unemployment, which yields a payoff of b. Firms use labor services L_j and energy E_j to produce a common consumption good c that is traded at $p \equiv 1$ across all local labor markets. Energy is produced outside of the model and sold at a price q, whereas firms post wages w_j to attract workers. Workers have heterogeneous preferences over firms' non-wage amenities and consume both energy and c.

Since workers' and firms' locations are fixed, and labor is hired locally, all decisions are taken within local labor markets, and I drop the subscript l for notational convenience in what

follows.

The Worker's Problem. Workers choose whether to work and, if so, which firm j to work for. In a second step, they decide how to allocate their income y between c and energy e. Let the utility of consumption be of the Stone-Geary type with a subsistence level of energy $\bar{e}$. The consumption problem then is:

$$\max_{c,e} \{\mu \ln(c) \ + \ (1-\mu)\ln(e-\bar{e})\} \quad \text{s.t. } c + q\,e \ \leq \ y_j, \quad y_j = \begin{cases} w_j, & j \in \{1,\ldots,J\}, \\ b, & j = 0, \end{cases}$$

which delivers the following indirect utility function:

$$V(y_j, q) = ln(y_j - q\bar{e}) + \underbrace{\mu ln\,(\mu) + (1-\mu) ln\left(\frac{1-\mu}{q}\right)}_{\equiv \Lambda(q)} \equiv V_j. \tag{4}$$

To link the theoretical and empirical analyses, I assume that $\bar{e}$ is a location l specific shifter that reflects fundamental differences in commuting infrastructure or housing. Labor markets with a higher $\bar{e}$ will have a higher expenditure share on energy, $S(y_j, q) = \frac{qe_j^*}{y_j} = (1-\mu) + \mu\frac{q\bar{e}}{y_j}$, and, therefore, suffer more severely from changes in q.

In line with modern monopsony models (e.g., Berger, Herkenhoff and Mongey, 2022; Card et al., 2018; Chan et al., 2024; Lamadon, Mogstad and Setzler, 2022; Volpe, 2024), workers value not only the indirect utility of consumption, V_j, but also non-pecuniary firm attributes. Their maximization problem is:

$$\max_{j \in \{0,1,\ldots,J\}} \{\beta_i V_j + \bar{\xi}_j + \xi_{ij}\}, \tag{5}$$

where β_i is drawn from some distribution $F(\beta)$ and governs the relative importance placed on consumption, $\bar{\xi}_j$ is the mean evaluation of firm j's non-pecuniary attributes among workers, and ξ_{ij} is an i.i.d. mean-zero individual-specific taste-shock. The quasi-linear preference specification implies that poorer workers value consumption relatively more than non-pecuniary choice attributes. In combination with the location-specific Stone-Geary utility shifter, it also implies that the relative ranking of firms will differentially depend on energy prices as a function of the labor market's exposure. While this creates differences in labor supply elasticities across markets, β_i introduces

within-market differences across firms.[15]

I normalize the average non-wage evaluation of unemployment to zero, i.e., $\bar{\xi}_0 = 0$. As highlighted by Chan et al. (2024), this normalization allows the identification of a utility scale by interpreting choices relative to the outside option of non-employment, i.e., net of the disutility of labor. Note that the model setup is general, so that $\bar{\xi}_j + \xi_{ij}$ can be interpreted as both explicitly provided firm amenities, e.g., flexible working hours, and implicit, unobserved attributes, e.g., work culture. In fact, with the standard assumption of extreme value distributed taste-shocks, it is easy to show that for any two firms $j \neq k$ that pay the same wage, $w_j = w_k$, and any β type, the following is true: $ln\left(\frac{Pr(j|w_j=w,\beta)}{Pr(k|w_k=w,\beta)}\right) = \bar{\xi}_j - \bar{\xi}_k$. That is, one interpretation of $\bar{\xi}_j$ is a non-pecuniary residual that rationalizes the observed firm size distribution.

Following the standard practice of modeling ξ_{ij} as drawn from an extreme value distribution (Gumbel), and denoting the vector of wages offered across all firms in the given local labor market by $\mathbf{W}$, the sorting probabilities of workers across firms are:

$$Pr(j|\beta, \mathbf{W}, q) = \frac{exp(\beta V_j + \bar{\xi}_j)}{\sum_{j'} exp(\beta V_{j'} + \bar{\xi}_{j'})}$$
$$Pr(j|\mathbf{W}, q) = \int Pr(j|\beta, \mathbf{W}, q) dF(\beta).$$

Taking everything together, the labor supply to firm j as a function of wages and prices is:

$$L_j(w_j|\mathbf{W}, q) = N \times Pr(j|\mathbf{W}, q). \tag{6}$$

The Firm's Problem. Firms produce with a firm-specific production technology, $f_j(L, E)$, that uses labor and energy as inputs and includes firm-specific TFP, z_j. They take energy prices and competitors' wages as given and optimally set wages to maximize profits, subject to the labor supply schedule derived in the worker's problem:

$$\max_{w_j, E_j} \{\Pi_j = f_j(L_j, E_j) - w_j L_j - qE_j \quad s.t. \quad L_j(w_j|\mathbf{W}, q) = N \times Pr(j|\mathbf{W}, q)\} \tag{7}$$

As is standard, I assume that firms cannot observe individuals' idiosyncratic tastes and therefore cannot price-discriminate among different workers. Given this classic monopsony setup,

[15] Volpe (2024) demonstrates that introducing heterogeneity across workers in β, i.e., in the relative weight placed on wages, creates richer and more realistic substitution patterns across firms. With heterogeneity in β, firms take into consideration which type of workers they can attract when offering a package of wages and non-wage attributes. This implies that labor supply elasticities differ across firms.

the offered wage will be a markdown on the firm's marginal revenue product. Define the labor supply elasticity as $\epsilon_j = \frac{\partial L_j(w_j)}{\partial w_j}\frac{w_j}{L_j}$, then we can re-arrange the firm's first-order-condition to obtain:

$$w_j = \frac{\epsilon_j}{1+\epsilon_j}\frac{\partial f_j(L_j, E_j)}{\partial L_j}. \tag{8}$$

Using the labor supply schedule, we can derive ϵ_j as:

$$\epsilon_j = \frac{w_j}{Pr(j)}\int \frac{\partial Pr(j|\beta)}{\partial w_j} dF(\beta) = \frac{w_j}{w_j - q\bar{e}} \underbrace{\int \beta \times \frac{Pr(j|\beta)(1-Pr(j|\beta))}{Pr(j)} dF(\beta)}_{\equiv \beta_j}, \tag{9}$$

where β_j is a firm-specific component that depends both on the composition and on the size of its workforce. Inserting this into (8) and re-arranging yields:

$$w_j = \frac{\beta_j}{1+\beta_j}\frac{\partial f_j(L_j, E_j)}{\partial L_j} + \frac{1}{1+\beta_j} q\bar{e}. \tag{10}$$

Wages in this setup are a weighted average of the firm's marginal revenue product and the minimum income level required for workers to subsist. If firms attract mainly high-β workers, i.e., workers that care more about consumption than they do about amenities, β_j will be higher, and firms' wage-setting is more closely aligned with their marginal revenue product. At the same time, for a given workforce composition, β_j is smaller in larger firms (L_j), giving them more market power.

Labor Market Equilibrium. Inserting (6) into (10) yields a system of $|L| \times |J_l|$ equations, the solution to which constitutes the labor market equilibrium vector of wages, such that, given q, they result from firms optimally posting wages to maximize profits and workers optimally sorting across firms to maximize their utility. Chan et al. (2024) show that in a setting with nested logit preferences on the worker side and a production technology with constant or decreasing returns to scale, the equilibrium exists and is unique.

5.2 Discussion of the Main Mechanism: Comparative Statics

There are some noteworthy aspects of the model's mechanics. First, from (9) one can show that, imposing some mild regularity conditions on $F(\beta)$ and considering cases without extreme dispersion in firm sizes and wages, the average (employment-weighted) labor supply elasticity

in a market increases in q, i.e, $\frac{\partial \bar{\epsilon}}{\partial q} > 0$.[16] For a given set of wages, workers lose disposable income in response to an energy price increase. This increases their marginal utility of consumption and, consequently, their labor supply elasticity. Internalizing this response, firms have an incentive to increase wages to attract workers, particularly in labor markets in which workers are more exposed to the increase in prices ($\bar{e}_l$). In fact, the second part of wage-setting equation (10) contains a direct pass-through of $\frac{\bar{e}_l}{1+\beta_j}$.

In the model, changes in energy prices trigger a reevaluation of the relative importance of non-pecuniary job components and consumption. While firms compensate workers for higher costs, the pass-through is only partial, leaving some workers insufficiently compensated. They, instead, are willing to accept (from their perspective) less valuable non-wage attributes and choose a firm they would not have chosen at a lower q. The impact of labor supply adjustments on the distribution of consumption is consequently not identical to its implication on welfare. Recently, Afrouzi et al. (2026) and Guerreiro et al. (2024) highlighted that workers dislike inflation because it requires them to take labor-market actions that are costly in utility terms. In my setting, such actions are quantified by the choice of lower-valued non-wage job attributes. The outlined theoretical framework generates clear predictions about workers' choices before and after energy cost increases and naturally lends itself as a tool to use workers' revealed preferences to inform a discussion of welfare costs that go beyond the pure average earnings responses of employees across local labor markets. The remainder of this paper proceeds by doing precisely that.

5.3 Model Estimation: Approach

The reduced-form empirical approach presented above compares the responses in job transitions and earnings to energy price changes across labor markets with different baseline energy expenditure shares. Identification relies on comparing otherwise identical labor markets. I replicate this idea in my theoretical framework by simulating a worker panel across a set of stylized labor markets that are identical in all respects except for $\bar{e}_l$. This creates heterogeneity in energy expenditures while holding worker preferences and production technologies fixed. I then re-estimate my main empirical specification on the simulated panel and use the regression coefficients, along with other

[16] In Appendix A.5.1, I provide a proof of this, including an explicit bound on the maximum dispersion of wages under which the relationship holds with certainty. Note that even if wages are too dispersed for $\frac{\partial \bar{\epsilon}}{\partial q} > 0$ to hold with monotonicity, the assumptions to prove that $\frac{\partial^2 \bar{\epsilon}}{\partial q \partial \bar{e}} > 0$ are considerably weaker. That is, the average labor supply elasticity in a given local labor market responds more positively (or less negatively) in markets where consumers spend a larger share of their budget on energy, which is precisely the idea underlying the empirical analysis of this paper. Appendix A.5.1 shows this in a corollary to the aforementioned proposition.

data moments, as targets in an indirect inference/simulated method of moments (SMM) exercise. This allows me to recover the distribution of workers' tastes for non-wage attributes that is needed to rationalize the observed data.

Since the model is static, I simulate a panel by solving for labor-market equilibrium at different values of q and tracking workers' choices. A plausible interpretation of this is that the simulated panel consists of steady-state comparisons. Given that the model contains no short-term adjustment frictions, the appropriate targets for estimation are the five-year comparisons presented above. I construct them by solving the model once in a baseline period, where $q = 1$, and then re-solving it for different draws of q' from the empirical distribution of five-year price changes. Appendix A.5.2 contains additional details on the construction of the worker panel.

The targeted reduced-form evidence is based on cross-market differences in energy cost shocks rather than on firm-specific shocks, which prevents the nonparametric recovery of non-wage attributes and marginal revenue products at the firm level. Instead, I impose the following assumptions for estimation:

1. Firms produce with a Cobb-Douglas production function: $f_j(L_j, E_j) = z_j \left[L_j^{1-\gamma_j} E_j^{\gamma_j} \right]^{\alpha}$, where α allows for the possibility of decreasing returns to scale, z_j is firm-specific TFP, and γ_j governs the energy-intensity of firm j's production.

2. The log of firm-productivity, $ln(z_j)$, and firms' average non-wage attributes, ξ_j, are jointly normal with means (μ_z, μ_ξ), standard deviations (σ_z, σ_ξ), and correlation $\rho_{z,\xi}$. $\rho_{z,\xi}$ is left unrestricted, which implies that higher productivity firms might also be desirable in non-pecuniary aspects. Firms' energy intensity is distributed according to a truncated log-normal with parameters $(\mu_\gamma, \sigma_\gamma)$ and support on the interval $(0, 1)$. TFP and the energy share are correlated with correlation coefficient $\rho_{z,\gamma}$.

3. Workers' relative preferences for consumption are distributed according to $ln(\beta) \sim \mathcal{N}(\mu_\beta, \sigma_\beta)$.

With these assumptions, the parameter vector to be recovered is:

$$\Theta = \{\underbrace{\mu, \bar{e}_l, \mu_\beta, \sigma_\beta, \mu_\xi, \sigma_\xi}_{\text{Worker Preferences}}, \underbrace{\mu_z, \sigma_z, \alpha, \mu_\gamma, \sigma_\gamma, \rho_{z,\gamma}}_{\text{Firm Technology}}, \rho_{z,\xi}, b\}.$$

I calibrate some parameters externally and estimate the remaining ones via SMM. For instance, $(\mu, \bar{e}_l)$ are calibrated directly to the observed energy expenditure shares in German

Table 8: Internally Estimated Parameters: Targets and Estimates

Parameter	Estimate	Interpretation	Data Moments
Worker Preferences:			
μ_β	1.334	Average Pref. Weight on Consumption	Reg. Coefficient of Δw on $S_c \frac{q'}{q}$
σ_β	0.142	Dispersion in Taste for Consumption	Reg. Coefficient of $Pr(E-E)$ on $S_c \frac{q'}{q}$
μ_ξ	-0.717	Disutility of Labor	Unemployment Rate
σ_ξ	0.490	Taste Dispersion of Non-Wage Attributes	Interquartile Range of Firm Size
Firm Production:			
μ_z	2.063	Average TFP	Mean(wage)
σ_z	1.295	TFP Dispersion	Var(wage)
σ_γ	0.941	Dispersion in Energy Intensity	Δ Switchers vs. Stayers in the Reg. Coeff. of Δw on $S_c \frac{q'}{q}$
Other:			
ρ	0.412	Corr. between Non-Wage Attributes and TFP	P90/P50 of Wages

counties and the empirical relationship between energy expenditure shares and total expenditures in EVS (see Appendix A.5.2 for details). I set $\alpha = 0.6$ as an intermediate value relative to the estimates in Chan et al. (2024) and Berger, Herkenhoff and Mongey (2022). The energy intensity parameter μ_γ is set to the average energy cost share of 3% among German manufacturing firms as documented by von Graevenitz and Rottner (2023). According to evidence from the same paper, there is a small but positive correlation between energy intensity and productivity, with $\rho_{z,\gamma} = 0.14$. b is chosen to match a 70% replacement rate of average wages in the data.

Table 8 provides an overview of the empirical moments that identify the remaining parameters. The mean and variance of firm productivity have obvious counterparts in the empirical mean and variance of wages. Since markdowns are also a function of the attractiveness of firms in non-pecuniary terms, $\rho_{z,\xi}$ is informative for the wage distribution: if highly attractive firms are also more productive, they have an increased wage-setting power, which depresses the right tail of the wage distribution. I thus discipline $\rho_{z,\xi}$ by targeting the 90th/50th percentile ratio of wages, in addition to the variance. The model implies that the remaining dispersion in firm size after adjusting for wage differences is driven by σ_ξ, such that I target the interquartile range of firm sizes. The normalization $\bar{\xi}_0 = 0$ implies that μ_ξ directly maps into the unemployment rate as it captures the average evaluation of firms' non-wage attributes, net of the disutility of labor.

The last set of parameters $(\mu_\beta, \sigma_\beta, \sigma_\gamma)$ directly relates to the reduced-form regression estimates presented above. As can be seen from the wage setting equation (10), β is a key determinant of the pass-through of energy prices into nominal earnings. The differential wage response across labor markets is thus informative of the average wage sensitivity μ_β. Since only a few workers switch in response to energy cost shocks, differences in E-E transitions are governed by the mass of workers in the tails of $F(\beta)$, i.e., σ_β. Finally, if the distribution of energy intensity across firms is

dispersed (high σ_γ), then the response of firms' marginal revenue product to changes in energy prices will be heterogeneous, which impacts their wage-setting (equation 10). Workers in high-energy-expenditure labor markets will respond more sensitively to this heterogeneity across firms, inducing additional selection among job switchers. The gap in pass-through between switchers and stayers thus helps to discipline σ_γ.

5.4 Model Estimation: Results and Welfare Analysis of Cost-of-Living Shocks

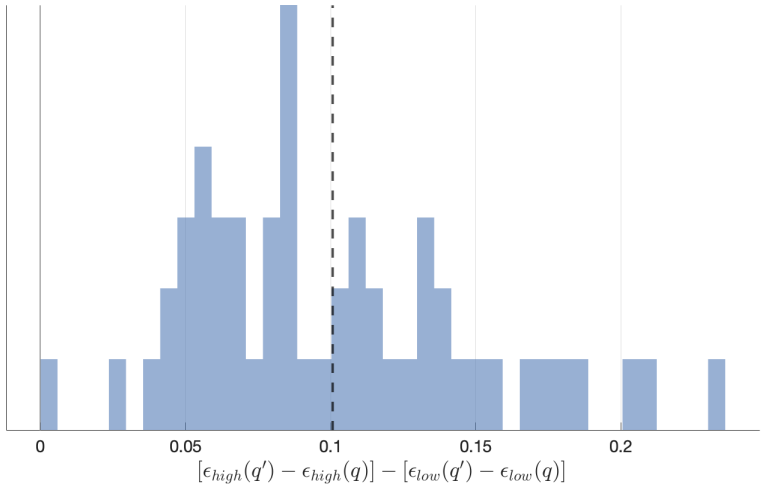
Table 8 reports the obtained parameter estimates from the simulated method of moments approach. The results imply substantial variation across firms in non-wage attributes (σ_ξ) and a positive correlation between these attributes and firm productivity ($\rho = 0.4$). That is, in line with findings from Lamadon, Mogstad and Setzler (2022), more productive firms also tend to have higher-valued non-wage attributes on average. Workers are also heterogeneous in their evaluation of consumption relative to non-pecuniary factors. The estimates imply that on average $\beta = 3.8$, with a standard deviation of about 0.5.[17]

By combining worker and firm heterogeneity, the model allows me to back out the implied own-wage labor-supply elasticity that firms face when setting wages. Given the rich interaction between the two types of heterogeneity in sorting, elasticities vary substantially across firms, with a mean employment-weighted value of 3.0 and an interquartile range of 1.6. By Equation (8), the average elasticity corresponds to a 25% average markdown of wages relative to the marginal revenue product of labor. This is at the upper end but in the range of estimates in previous studies, implying substantial labor market power on the firm side (Azar, Berry and Marinescu, 2022; Berger, Herkenhoff and Mongey, 2022; Chan et al., 2024; Kroft et al., 2025; Lamadon, Mogstad and Setzler, 2022; Volpe, 2024; Yeh, Macaluso and Hershbein, 2022).

In the spirit of the reduced-form empirical analysis of this paper, Figure 4 shows a difference-in-difference thought experiment. It illustrates how firm-level labor supply elasticities respond differently across markets when moving from a baseline period with $q = 1$ to $q' = 1.2$. This corresponds to the median empirical price change across the 5-year windows considered. The contrast is between the highest energy exposure decile and the lowest. Across all firms, the labor supply elasticity responds more positively in markets where workers face a larger cost shock due to greater exposure to energy price changes. Since there is an immediate mapping from labor

[17] Since β is log-normal, $\mathbb{E}[\beta] = exp(\mu_\beta + \sigma_\beta^2/2)$ and $SD(\beta) = \sqrt{[exp(\sigma_\beta^2) - 1]exp(2\mu_\beta + \sigma_\beta^2)}$.

Figure 4: Differene-in-Difference of Labor Supply Elasticities



Note: The Figure shows the difference in changes of estimated firm-level own-wage labor supply elasticities when energy prices increase from $q = 1$ to $q' = 1.2$ between high and low-exposure markets. The dashed vertical line indicates the mean. See equation 9 on how elasticities are calculated.

Table 9: The Effect of Energy Cost Shocks in the Data and the Model

	Pr(E-to-E)	$\Delta ln(earnings)$		
	(1)	(2)	(3)	(4)
Energy Cost Shocks in the Data	0.330	0.804	1.258	0.398
	(0.193)	(0.125)	(0.218)	(0.101)
Energy Cost Shocks in the Model	0.319	0.780	1.253	0.393
Sample	Full	Full	Switchers	Stayers

supply elasticities to wages, Figure 4 directly highlights the key mechanism: workers can partially recover their earnings losses through a differential targeting of firms. In line with this, the targeted model-implied pass-through of energy prices into earnings, as well as the pass-through conditional on switching firms and staying with the current employer, respectively, match their empirical counterparts closely (see Table 9; Appendix A.5.2 further discusses model fit relative to the data).

Changes in Welfare across Labor Markets. A key advantage of the structural model is that it allows me to measure welfare changes in consumption-equivalent units and to decompose them into the effect of changes in nominal earnings and the non-pecuniary channel arising because of

worker sorting. The expected utility of a type β worker in labor market l at energy price q is:

$$
\begin{aligned}
EU_l(q|\beta) &= \tilde{\gamma} + ln\left(\sum_j exp\{\beta V_j(w_j(q), q) + \bar{\xi}\}\right) \\
&= \underbrace{\sum_j Pr(j|q,\beta)\beta V_j(w_j(q), q)}_{\text{Consumption}} + \underbrace{\sum_j Pr(j|q,\beta)\left[\bar{\xi}_j - ln(Pr(j|q,\beta))\right]}_{\text{Non-Pecuniary}} + \tilde{\gamma},
\end{aligned}
$$

where the second line follows from a standard decomposition of the log-sum.[18] The first term measures the systematic consumption component, whereas the latter part quantifies non-pecuniary costs, consisting of systemic tastes ($\bar{\xi}_j$) and idiosyncratic gains from joining firms (ξ_{ij}); $\tilde{\gamma}$ is the Euler-Mascheroni constant.

Now consider an increase in energy prices from q to q' and define the consumption equivalent, τ^{CE}, as the lump-sum transfer in daily earnings that makes workers indifferent in expectation between both states of the world: $0 = \mathbb{E}_\beta[EU_l(q'|\tau^{CE}, \beta) - EU_l(q|\beta)]$. To a first-order approximation:

$$
\tau^{CE} \approx -\frac{\mathbb{E}_\beta[EU_l(q'|\beta) - EU_l(q|\beta)]}{\mathbb{E}_\beta\left[\beta \sum_j Pr(j|q',\beta)\frac{1}{w_j(q') - q'\bar{e}_l}\right]}.
$$

Replacing the numerator with the difference between states in consumption, as well as the systematic and idiosyncratic components of taste, allows for a decomposition of the total welfare effect. Since the empirical reduced-form coefficients that discipline the structural estimation are differential responses to energy prices, I treat the model results similarly and present differences in welfare costs across markets with varying energy-price exposure.

Table 10 contains estimates of relative welfare costs for labor markets at different deciles of the energy expenditure distribution. As expected, the consumption equivalent increases with energy expenditures. Contrasting labor markets in the highest with those in the lowest decile of energy shares reveals an additional welfare cost of 66 cents ($\approx$ 0.6%) in daily earnings for a median energy price change. The next two columns decompose this average effect into a pecuniary and non-pecuniary channel. Although increases in nominal earnings offset about 80% of real wage losses, the pass-through is incomplete. Taken together with the distaste for the change in

[18] Denote the sum inside the logarithm by Z. Workers choice probabilities then are $Pr(j|\beta) = exp(\beta V_j + \bar{\xi}_j)/Z \iff ln(Pr(j|\beta)) = \beta V_j + \bar{\xi}_j - ln(Z)$. Multiply by $Pr(j|\beta)$, sum over j, and rearrange: $ln(Z) = \sum_j Pr(j|\beta)(\beta V_j + \bar{\xi}_j) - \sum_j Pr(j|\beta) ln(Pr(j|\beta))$.

Table 10: Relative welfare costs across exposure deciles (CE in Daily Euros)

	τ^{CE}	Decomposition: $\tau^{CE}_{\text{consump.}}$	$\tau^{CE}_{\text{non-pecun.}}$
Decile 5 - Decile 1	0.32	0.22	0.10
Decile 10 - Decile 5	0.34	0.24	0.10
Decile 10 - Decile 1	0.66	0.45	0.21

Notes: Welfare changes are computed in consumption-equivalent euros by equating county-level expected utility before and after an increase in energy prices by 20% (median increase over 5-year window). The decomposition is exact: $\tau^{CE} = \tau^{CE}_{\text{consump.}} + \tau^{CE}_{\text{non-pecun.}}$.

relative prices, this yields relevant welfare costs for workers in consumption terms ($\tau^{CE}_{\text{consump.}}$), amounting to roughly two-thirds of the cross-labor-market gradient in welfare costs. The remaining third comes from non-pecuniary costs. Workers in markets with greater energy-shock exposure take costly actions to offset relatively larger cost increases, thereby exacerbating the dispersion of welfare among consumers. This is despite the positive correlation between a firm's systematic non-pecuniary attributes and its productivity ($\rho_{z,\xi} > 0$). Workers respond to energy cost shocks by switching to higher-paying firms with lower average and idiosyncratic non-pecuniary benefits. From their individual perspective, they choose less appealing firms when energy prices increase, which is not surprising since firms that were initially dominated in both wage and non-wage terms would not have been selected to begin with. Overall, the results clearly indicate that, even though adjustments in nominal earnings offset the distributional consequences of energy cost shocks, they do not eliminate them completely. Residual impacts remain both on the consumption side and through higher non-pecuniary welfare costs.

6 Concluding Remarks

This paper estimates the pass-through of cost-of-living shocks into earnings and demonstrates that individual labor supply responses significantly reduce the distributional consequences of price changes. Relying on county-level differences in consumption patterns to construct measures for heterogeneous exposure to movements in energy prices, I demonstrate that workers with higher exposure partially offset real income losses through both job-to-job mobility and on-the-job bargaining. A one-standard-deviation higher local energy cost shock raises annual nominal earnings growth by about 0.4 percentage points, covering 30–35% of the additional cost in the short run and close to 80% over a five-year horizon. These results highlight that consumption-side shocks are transmitted to the labor market through worker responses, even when firms face similar

non-labor input costs.

The structural framework presented in the paper formalizes an explanation by showing how cost shocks increase workers' marginal utility of income and, in turn, the elasticity of labor supply that firms face. This generates both wage increases for job stayers and positive selection of transitions for switchers. A welfare implication is that even when real wage losses are partially avoided, workers must sacrifice non-wage amenities to do so, highlighting the hidden costs of inflation that are not visible in earnings alone. A question for future research is whether such cost-shock-triggered trade-offs yield one-off earnings gains or permanently place workers in careers with steeper earnings growth trajectories. In either case, cost-of-living shocks can shape workers' labor market trajectories and, consequently, their lifetime earnings.

While not the primary focus of this paper, the results presented here also offer insights into policies to address the distributional consequences of cost shocks. First, earnings adjustments buffer much of the shock, but they are incomplete and heterogeneous. Younger workers and those with higher education recover quickly, while older workers and job stayers without a university degree do not. This suggests that targeted policies might be better suited to smooth distributional shocks than broad-based subsidies to energy consumption. Second, the finding that higher energy costs increase the intensity and selectivity of job mobility underscores the importance of labor market institutions, such as worker councils, and mobility frictions in shaping how cost shocks translate into inequality. Finally, the results suggest that cost-of-living shocks can feed into wages, albeit to a limited extent and as a consequence of workers re-sorting across firms. This is informative for debates about the extent of price-wage spirals (Darougheh et al., 2025) and for how to optimally structure monetary policy responses to cost-of-living (Olivi, Sterk and Xhani, 2023) or energy cost shocks (Bobasu, Dobrew and Repele, 2025; Gnocato, 2025).

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Appendix

Labor Market Dynamics after Cost-of-Living Shocks

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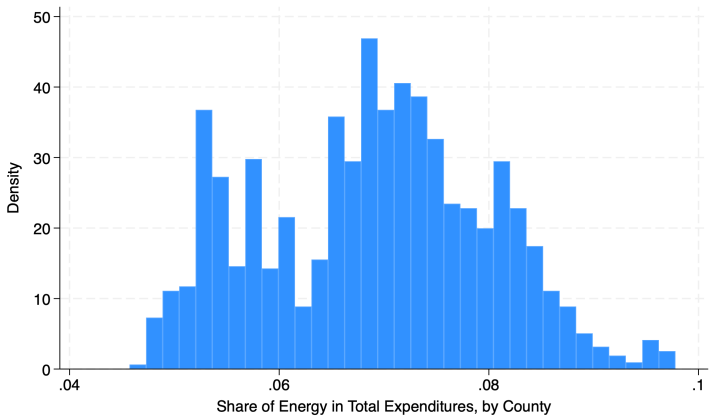
April 2026

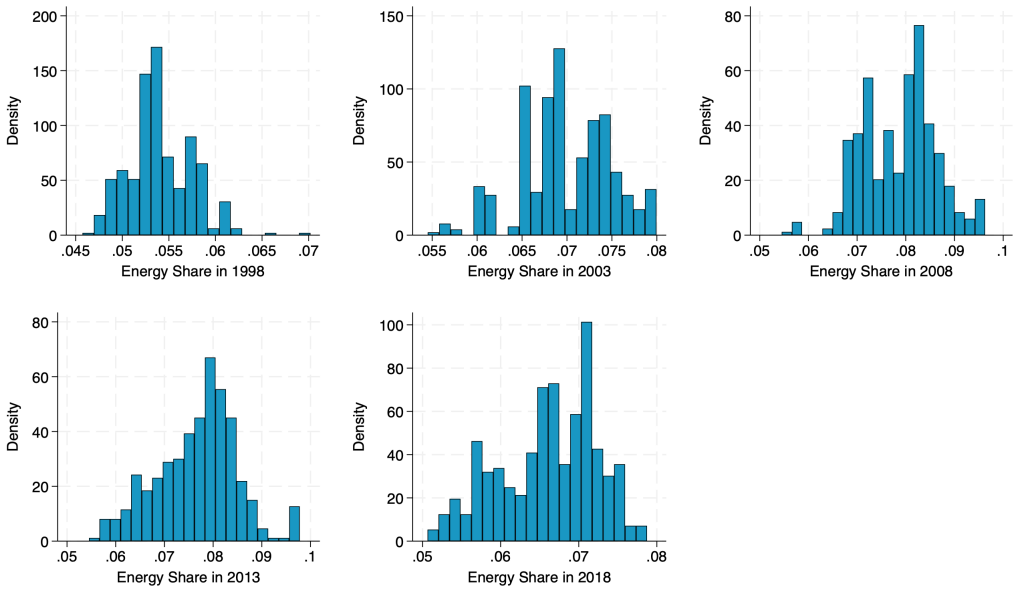
A.1 Additional Figures

Figure A1: Estimated Energy Expenditure Shares by County



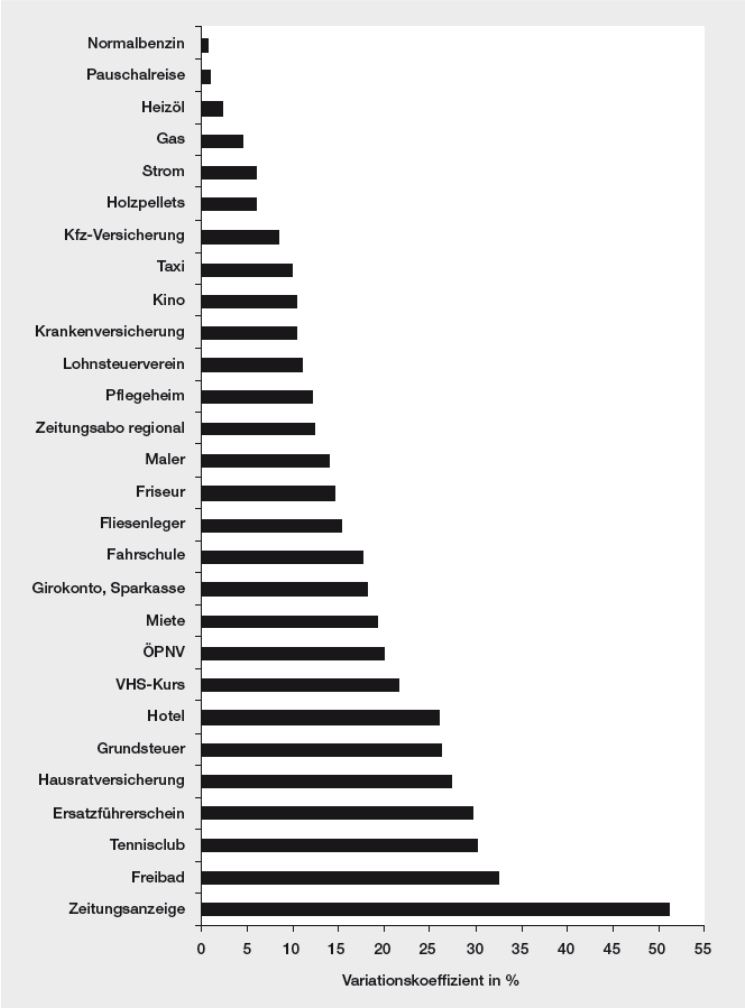
Note: The Figure shows the distribution of estimated county-level averages for total expenditure shares of energy goods, including gasoline, gas, oil, and other types of energy. Source: Einkommens- und Verbrauchsstudie 1998, 2003, 2008, 2013, and 2018, each Basic File 3, own calculations. For a year-by-year breakdown, see Figure A2.

Figure A2: Distribution of Estimated Energy Expenditure Shares, by EVS Wave



Note: The Figure shows the distribution of estimated county-level averages for total expenditure shares of energy goods, including gasoline, gas, oil, and other types of energy. Source: Einkommens- und Verbrauchsstudie 1998, 2003, 2008, 2013, and 2018, each Basic File 3, own calculations.

Figure A3: Variation in Prices across Counties, by Type of Good



Note: The figure is taken from BBSR (2009) and displays coefficients of variation for prices of various goods across German counties in 2008. The translation of each good type from German to English is (in decreasing order): Gasoline, package holidays, heating oil, gas, electricity, wood pellets, car insurance, taxi fee, cinema ticket, health insurance, membership fee income tax association, nursing home, subscription to regional newspaper, painter, haircut, tiler, driving instructions, fees for checking account, rent, public transportation, community college course fees, hotel, property tax, household insurance, replacement driving license, membership fee tennis club, outdoor pool, newspaper advertisement.

A.2 Additional Tables

Table A1: Correlation Coefficients of Expenditure Shares with County Characteristics

	Population	Population Density	Commuter Share
Energy Exp. Share	-0.184	-0.494	0.119
	(0.000)	(0.000)	(0.011)
	New Housing	Access to Public Transport	Dist. to Regional Center
Energy Exp. Share	-0.190	-0.448	0.371
	(0.000)	(0.000)	(0.000)

Notes: The Table displays correlation coefficients between county-level expenditure shares in a given year and other county characteristics obtained from the INKAR database at BBSR. P-values are displayed in parentheses. Commuter share is the share of all employees who commute $>$ 50km. Access to Public Transport is the share of inhabitants living within a 1km radius of a public transport stop offering at least 20 rides a day. New Housing is the fraction of newly built housing units per 1,000 existing housing units. Distance to Regional Center measures the time, in minutes, it would take an average inhabitant to reach a regional center (*Oberzentrum*) by car.

Table A2: Year-to-Year Energy Cost Shocks and Earnings: Alternative Models

	$Outcome = ln(Earnings_{it}/Earnings_{i,t-1})$				
	(1)	(2)	(3)	(4)	(5)
Energy Cost Shock	0.544***	0.387***	0.339***	0.721***	0.267***
	(0.078)	(0.050)	(0.056)	(0.138)	(0.058)
	Carbon IV	Top-Coded	Region-Year FE	Fixed $S^g_{c,0}$	Within-Firm
N Individuals	639,056	639,076	639,056	639,056	516,762
N Total	7,101,803	7,116,270	7,101,803	7,101,803	4,666,710

Notes: The Table presents alternative models of the relationship between energy cost shocks and earnings. Column (1) instruments the energy cost shock from Table 3 with changes in European carbon prices taken from Känzig (2023) and presents 2SLS results. The instrument is $\left[\sum_g S^g_{c,t-1}\right] \times CPSurprise_t$, where $CPSurprise_t$ is the yearly sum of monthly carbon price changes relative to consumer prices for electricity. Column (2) re-estimates τ in specification 1, using non-imputed/top-coded earnings as the outcome variable. Column (3) additionally conditions on year-by-macro region (*Bundesland*) fixed effects. Column (4) fixes the estimated energy expenditure shares at the earliest available level (1998). Column (5) compares individuals working in the same establishment in period $t-1$. All specifications include individual and county fixed effects, as well as year-by-sector by within-sector firm-size tercile fixed effects. Each model additionally allows yearly fixed effects to vary over: quintiles of the county-sector-level distribution of employment concentration (Hermann-Herfindahl-Index), quintiles of the energy intensity of local production, measured as the share of establishments in each county that are operating in energy-intensive sectors according to Känzig (2023) (i.e., agriculture, utilities, transportation, manufacturing), as well as across differences in the lagged county-level unemployment rate. Standard errors are clustered at the county level.

Table A3: The Effect of Year-to-Year Energy Cost Shocks on Earnings in LLMs

	$Outcome = ln(Earnings_{it}/Earnings_{i,t-1})$				
	(1)	(2)	(3)	(4)	(5)
Yearly Energy Cost Shock	0.444***	0.426***	0.394***	0.417***	0.349***
	(0.069)	(0.064)	(0.060)	(0.075)	(0.059)
N Individuals	639,056	639,056	639,056	639,056	639,056
N Total	7,101,803	7,101,803	7,101,803	7,101,803	7,101,803
Match HHI		✓			✓
Match Energy-Intensity			✓		✓
Match Unemployment				✓	✓

Notes: The Table is identical to Table 3, replacing all shocks and variables measured at the county-level with respective measurements at the commuting-zone level. The crosswalk from counties to commuting zones is based on the BBSR definition. There are 223 unique commuting zones, and standard errors are clustered at that level.

Table A4: Year-to-Year Energy Cost Shocks on Earnings; by Type

		Exclude Energy Type x:			
		Gasoline	Gas	Oil	Other
	(1)	(2)	(3)	(4)	(5)
$\sum_{g \neq x} S^{g}_{c,t-1} \frac{P^{g}_{t}}{P^{g}_{t-1}}$		0.412***	0.314***	0.408***	0.216***
		(0.060)	(0.051)	(0.061)	(0.045)
$S^{gasoline}_{c,t-1} \frac{P^{gasoline}_{t}}{P^{gasoline}_{t-1}}$	0.336***				
	(0.076)				
$S^{gas}_{c,t-1} \frac{P^{gas}_{t}}{P^{gas}_{t-1}}$	0.454***				
	(0.100)				
$S^{oil}_{c,t-1} \frac{P^{oil}_{t}}{P^{oil}_{t-1}}$	0.031				
	(0.082)				
$S^{other}_{c,t-1} \frac{P^{other}_{t}}{P^{other}_{t-1}}$	0.679***				
	(0.146)				
N Individuals	639,056	639,056	639,056	639,056	639,056
N Total	7,101,803	7,101,803	7,101,803	7,101,803	7,101,803

Notes: The Table displays estimates for τ in specification 1, proxying individual energy-price inflation with county-level averages as estimated in Section 3.2. All specifications include individual and county fixed effects, as well as year-by-sector by within-sector firm-size tercile fixed effects. Each model additionally allows yearly fixed effects to vary over: quintiles of the county-sector-level distribution of employment concentration (Herfindahl-Hirschmann-Index), quintiles of the energy intensity of local production, measured as the share of establishments in each county that are operating in energy-intensive sectors (i.e., agriculture, utilities, transportation, manufacturing), as well as across differences in the lagged county-level unemployment rate. Standard errors are clustered at the county level.

Table A5: Pass-Through When Dropping Counties with High (Low) Share of Employment in Sectors with High Variance in Energy-Intensity

	$Outcome = ln(Earnings_{it}/Earnings_{i,t-1})$			
	(1)	(2)	(3)	(4)
Energy Cost Shock	0.304***	0.284***	0.281***	0.334***
	(0.046)	(0.046)	(0.049)	(0.044)
N Individuals	639,056	590,676	543,696	555,170
N Total	7,101,803	6,450,972	5,840,294	5,971,931
Sample	Full	Drop Top 10%	Drop Top 20%	Drop Bottom & Top 10%

Notes: The Table displays estimates for τ in specification 1, proxying individual energy-price inflation with county-level averages as estimated in Section 3.2: $C^1_{ct} = \sum_g S^g_{c,t-1} \frac{P^g_t}{P^g_{t-1}}$. Each specification includes individual and county fixed effects, as well as year-by-sector by within-sector firm-size tercile fixed effects. Each model additionally allows yearly fixed effects to vary over: quintiles of the county-sector-level distribution of employment concentration (Herfindahl-Hirschmann-Index), quintiles of the energy intensity of local production, measured as the share of establishments in each county that are operating in energy-intensive sectors (i.e., agriculture, utilities, transportation, manufacturing), as well as across differences in the lagged county-level unemployment rate. Firm size is measured by the number of employees in the firm that individual i worked for in year $t-1$. Columns (2), (3), and (4) drop all individuals residing in counties for which the employment share in sectors with a known high variance of energy intensity is among the top 10%, top 20%, and Bottom or Top 10% of the cross-county distribution, respectively. The mean employment shares of high-energy-intensity-variance industries per decile (40 counties) of the cross-county distribution are: 0.25%, 0.6%, 1.0%, 1.4%, 1.8%, 2.4%, 3.2%, 4.2%, 6.6%, and 15.0%. The 3-digit sectors with the highest within-sector variance in energy-intensity according to Petrick, Rehdanz and Wagner (2011) are: (i) mining, (ii) glass, (iii) coke, petroleum products, and nuclear fuel, (iv) locomotives and rolling stock, (v) non-metal recycling, (vi) basic and agro-chemicals, (vii) metal recycling, (viii) basic iron and steel, tubes, (ix) veneer sheets, plywood, laminboard, and the like, and (x) builders' carpentry and joinery. Energy intensity is defined as the kWh of energy input per €1,000 of output. The measure of variation in energy intensity is (p90-p10)/p50. Standard errors are clustered at the county level.

Table A6: The Effect of Cumulative Energy Cost Shocks on Earnings; Different Horizons

	$Outcome = ln(Earnings_{it}/Earnings_{i,t-s})$				
	$s=1$	$s=2$	$s=3$	$s=4$	$s=5$
Energy Cost Shock	0.304***	0.566***	0.649***	0.731***	0.804***
	(0.046)	(0.072)	(0.094)	(0.108)	(0.125)
N Individuals	639,056	621,628	601,770	546,337	493,397
N Total	7,101,803	6,319,090	5,582,994	4,854,828	4,204,432

Notes: The Table displays estimates for τ in specification 1, proxying individual energy-price inflation with county-level averages as estimated in Section 3.2: $C^s_{ct} = \sum_g S^g_{c,t-1} \frac{P^g_t}{P^g_{t-s}}$. Each specification includes individual and county fixed effects, as well as year-by-sector by within-sector firm-size tercile fixed effects. The models additionally allow yearly fixed effects to vary over: quintiles of the county-sector-level distribution of employment concentration (Herfindahl-Hirschmann-Index), quintiles of the energy intensity of local production, measured as the share of establishments in each county that are operating in energy-intensive sectors (i.e., agriculture, utilities, transportation, manufacturing), as well as across differences in the lagged county-level unemployment rate. Firm size is measured by the number of employees in the firm that individual i worked for in year $t-s$. Standard errors are clustered at the county level.

Table A7: The Effect of Cumulative Energy Cost Shocks on Earnings and Job Mobility

Outcome:	$s=1$	$s=2$	$s=3$	$s=4$	$s=5$
$Pr(E-E)$	0.271*	0.457**	0.451**	0.381*	0.330*
	(0.161)	(0.187)	(0.194)	(0.197)	(0.193)
$\Delta(earnings)\|switching$	0.523***	0.960***	1.064***	1.168***	1.258***
	(0.125)	(0.163)	(0.195)	(0.203)	(0.218)
$\Delta(earnings)\|staying$	0.277***	0.345***	0.330***	0.377***	0.398***
	(0.039)	(0.060)	(0.079)	(0.089)	(0.101)

Notes: The Table displays estimates for τ in specification 1, proxying individual energy-price inflation with county-level averages as estimated in Section 3.2: $C^s_{ct} = \sum_g S^g_{c,t-1} \frac{P^g_t}{P^g_{t-s}}$. Each specification includes individual and county fixed effects, as well as year-by-sector by within-sector firm-size tercile fixed effects. The models additionally allow yearly fixed effects to vary over: quintiles of the county-sector-level distribution of employment concentration (Herfindahl-Hirschmann-Index), quintiles of the energy intensity of local production, measured as the share of establishments in each county that are operating in energy-intensive sectors (i.e., agriculture, utilities, transportation, manufacturing), as well as across differences in the lagged county-level unemployment rate. Firm size is measured by the number of employees in the firm that individual i worked for in year $t-s$. Standard errors are clustered at the county level.

A.3 Imputation of County-Level Expenditure Shares

Both data sets provided by IAB-FDZ, the Sample of Integrated Employer-Employee Data and the Establishment History Panel, contain geographic information on firms' and individuals' locations at the level of the 400 administrative counties in Germany. The Income- and Expenditure Survey (EVS) data set, on the other hand, provides information on Germany's 16 regions as the smallest level of geographical aggregation. It does, however, further distinguish between households living in areas corresponding to one of three agglomeration types depending on the population density and share of the local population that lives in cities: urban areas ($> 50\%$ lives in cities and population density is above $150/km^2$), semi-urban ($> 50\%$ lives in cities, population density is below $150/km^2$ or $< 50\%$ lives in cities but population density is above $100/km^2$), and rural areas ($< 50\%$ lives in cities and population density is below $100/km^2$). The basis for this assessment is the household's county of residence.

To harmonize the geographic units in all data sets and to predict county-level energy expenditure shares as suggested in Section 3.2, I first use region-by-agglomeration type locations, c, to estimate regressions of the type

$$S^g_{h,c} = \pi_c + X_{h,c}\beta + u_{h,c}, \tag{11}$$

where $S^g_{h,c}$ is household h's expenditure share on energy good g, π_c are location c fixed effects and $X_{h,c}$ is a set of covariates.

There is a direct mapping between each German administrative county and c, which allows me to predict stylized county-level expenditure shares. To further distinguish between counties within a region-by-agglomeration type cell, I use information on county-level characteristics provided by the Federal Institute for Research on Building, Urban Affairs, and Spatial Development (BBSR) and extract information on each county's infrastructural features that likely predict energy expenditures from the INKAR database (https://www.inkar.de/): average living space per household member and the share of buildings that consist of at least three apartments as proxies for the available housing stock, and the average number of cars per inhabitant as a proxy for commuting patterns. Each variable has an empirical counterpart in the EVS data that is included as a covariate in $X_{h,c}$. After estimating the parameters of specification (11), I predict county-level expenditure shares using $\hat{\pi}_c$ and the estimated coefficients on each of the three outlined variables. All other socio-demographic information is kept at the sample mean to adjust for household sorting as

discussed in Section 3.2.

Table A8 contains point estimates of socio-demographic predictors of total energy expenditures for each survey wave (1998, 2003, 2008, 2013, 2018). The regressions additionally adjust for quarter-of-year fixed effects to filter out seasonal trends in energy expenditures, as well as for region-by-agglomeration-type interactions, as discussed above. As expected, energy expenditure shares are decreasing in gross income but increasing in the available living space. In all five survey waves considered, energy expenditures are also strongly and positively correlated with the number of cars in the household. Households in which the household head has a university degree or is female, on the other hand, spend less on energy. The labor force status that most consistently predicts higher energy expenditure shares is unemployment.

Table A8: Predictors of Energy Expenditure Shares – EVS Data

Sample Year:	1998	2003	2008	2013	2018
	(1)	(2)	(3)	(4)	(5)
HH Head has/is:					
Married (=1)	0.001	0.003***	0.006***	0.003***	0.000
	(0.000)	(0.001)	(0.001)	(0.001)	(0.001)
University Degree (=1)	-0.003***	-0.004***	-0.005***	-0.005***	-0.006***
	(0.000)	(0.0000)	(0.001)	(0.002)	(0.000)
Female (=1)	-0.002***	-0.004***	-0.002***	-0.003***	-0.003***
	(0.000)	(0.001)	(0.001)	(0.001)	(0.000)
German (=1)	-0.003***	0.004**	0.004**	-0.000	0.001
	(0.001)	(0.002)	(0.002)	(0.002)	(0.001)
In Part-Time Employment (=1)		0.002*	0.001	0.002***	0.002***
		(0.001)	(0.001)	(0.001)	(0.001)
HH Head's Labor Force Status:					
Self-Employed	0.004*	-0.005	0.002	-0.003	0.009
	(0.002)	(0.004)	(0.005)	(0.007)	(0.006)
Civil Servant	0.003	-0.005	-0.002	-0.007	0.002
	(0.002)	(0003)	(0.005)	(0.007)	(0.006)
Employee	0.006***	0.001	0.007	-0.004	0.006
	(0.002)	(0.003)	(0.005)	(0.007)	(0.006)
Unemployed	0.010***	0.009**	0.022***	0.012*	0.028***
	(0.002)	(0.004)	(0.005)	(0.007)	(0.006)
Retiree	0.003	0.001	0.006	-0.004	0.007
	(0.002)	(0.004)	(0.005)	(0.007)	(0.006)
Pensioner	-0.001	-0.005	-0.001	-0.006	0.000
	(0.002)	(0.004)	(0.005)	(0.007)	(0.006)
Student	-0.005*	-0.006	-0.006	-0.026***	-0.009
	(0.003)	(0.004)	(0.005)	(0.007)	(0.006)
Other	-0.006**	0.002	0.007	0.003	0.011
	(0.003)	(0.004)	(0.005)	(0.008)	(0.007)
Age	-0.000***	-0.000***	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
HH Size	0.006***	0.003***	0.009***	0.007***	0.006***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
N of Employed HH Members	0.003***	0.005***	0.005***	0.005***	0.002***
	(0.000)	(0.000)	(0.001)	(0.001)	(0.000)
Total Exp. (in 1,000)	-0.001***	-0.003***	-0.003***	-0.003***	-0.003***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Equival. Income (in 1,000)	-0.049***	-0.059***	-0.116***	-0.113***	-0.020***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Living Space (in 100 m^2)	0.005***	0.016***	0.017***	0.005***	-0.002
	(0.000)	(0.000)	(0.000)	(0.000)	(0.002)
N of Cars in HH	0.027***	0.002***	0.037***	0.032***	0.026***
	(0.001)	(0.000)	(0.001)	(0.001)	(0.001)
House not Owned / Renter (=1)	0.002***	-0.003***	-0.008***	-0.001*	0.008***
	(0.000)	(0.001)	(0.001)	(0.001)	(0.001)
Appartment in 3+ Unit Building (=1)				-0.018***	-0.013***
				(0.001)	(0.001)
N	49,720	42,744	55,110	53,490	52,782
R^2	0.230	0.150	0.162	0.177	0.206

Notes: The table presents point estimates of energy expenditure shares ($\sum_g S^g_{h,t}$) on characteristics of the household and household head. Each regression additionally adjusts for quarter-of-year fixed effects and region-by-agglomeration-type fixed effects, as discussed above. The reference category for labor force status is self-employment in the agricultural sector/farmers. Source: Einkommens- und Verbrauchsstudie 1998, 2003, 2008, 2013, and 2018, each Basic File 3, own calculations.

A.4 Heterogeneity in the Response to Cost-of-Living Shocks

The presented empirical evidence in this paper demonstrates that average nominal earnings respond positively to changes in county- and energy-specific CPIs. This allows consumers to recover a non-negligible share of predicted real consumption losses already in the short term, particularly if the cost shocks encourage job-to-job mobility. This section explores treatment effect heterogeneity and whether specific subgroups are particularly able to profit from adjustments in their labor supply behavior.

Before turning to heterogeneity based on worker characteristics, however, Table A9 considers heterogeneity over time. The paper reports average results from the pool of observations over the period 1999 to 2019, yielding an estimated average recovery of real earnings losses of around 31%. Columns (1) through (4) indicate that this average masks a substantial heterogeneity. In the early 2000s, a period when Germany was characterized as the "sick man of Europe" (Dustmann et al., 2014), in part due to its inflexible labor market, the pass-through was considerably smaller. To a lesser extent, this is also true for the period 2010-2014, which contains the European sovereign debt crisis. After 2020, the point estimates increase again, though estimated with considerably greater uncertainty, as indicated by standard errors of around 0.28. This is intuitively appealing, as the period after 2020 combines two macro events that influenced the German labor market: the COVID-19 pandemic, which significantly altered labor market dynamics, and the war in Ukraine, which led to an unprecedented rise in energy prices (at least over the sample period considered).

The combination of the two events makes it difficult to interpret the relative magnitude of pass-through compared with earlier periods. On the one hand, the COVID-19 pandemic led to

Table A9: The Effect of Energy Cost Shocks on Earnings, by Time Period

	$Outcome = ln(earnings_{it}/earnings_{i,t-1})$				
	(1)	(2)	(3)	(4)	(5)
Yearly Cost Shock	0.160*	0.807***	0.356***	0.388***	0.767***
	(0.083)	(0.124)	(0.130)	(0.123)	(0.210)
N Individuals	416,565	422,411	397,257	375,940	295,773
N Total	1,881,576	1,805,100	1,680,223	1,593,958	1,073,788
Time Period	2000-2004	2005-2009	2010-2014	2015-2019	2020-2023

Notes: $^{*}p < 0.1,^{**}p < 0.05,^{***}p < 0.01$. The Table displays estimates for τ in specification 1, proxying individual energy-price inflation with county-level averages as estimated in Section 3.2: $C^s_{ct} = \sum_g S^g_{c,t-1} \frac{P^g_t}{P^g_{t-s}}$. Covariates are identical to those in Table 3. Standard errors are clustered at the county level.

Table A10: The Effect of Cost Shocks on Earnings, by Age

Outcome	Age 25-40	Age 41-55
$\Delta ln(earnings), yearly$	0.460***	0.130***
	(0.074)	(0.040)
N Individuals	455,499	412,248
N Total	3,365,942	3,696,842

Notes: The Table displays estimates for τ in specification 1, estimated separately by subsample on the outcomes displayed in column 1. Covariates are identical to those in Table 3. Standard errors are clustered at the county level and displayed in parentheses.

a strong uptake of remote work arrangements that persists to some extent in the post-pandemic period. Bagga et al. (2025) use this fact as motivating evidence to show that such shocks to the "amenity values" of firms can induce significant labor reallocation towards firms that offer the option to work from home. In the theoretical framework I discuss in Section 5, the pass-through of energy prices into nominal earnings is limited in such a case because firms compete less intensively on the wage dimension and more intensively on amenities. On the other hand, in my theoretical framework, the labor supply elasticity of workers depends on the marginal utility of income. With a concave utility over consumption, larger shocks to their real earnings (larger increases in energy prices) generate larger marginal utility responses. Both macro forces induced by the COVID-19 pandemic and the high energy prices in the aftermath of the war in Ukraine consequently interact, making the post-2020 period relatively unique. While this is an interesting period to consider, the paper focuses on the period up to 2019 to abstract from such complications.

Moving on to heterogeneity across worker types, Table A10 shows that the pass-through of cost shocks into earnings decreases with age. Employees aged 25 to 40 can recover 46% of county-specific cost increases within a year, whereas the corresponding figure for the age group 41 to 55 is 13%. Given the critical role of job mobility discussed in the paper and the generally higher mobility among younger workers, the negative age gradient is unsurprising. In an average year, the probability of observing an E-E transition among relatively younger workers is 23% compared to a 14% probability among workers aged 41 to 55.

At the same time, it is unlikely that job mobility is a uniformly effective strategy across the population. In recent work, Kline (2025) highlights the central role of the distribution of workers' outside options for determining firms' labor market power and, in turn, the pass-through of shocks into earnings. On average, people in higher-exposure counties are more likely to change

Table A11: The Effect of Cost Shocks on Earnings, by Employment Concentration

Outcome	Quintile 1	Q2	Q3	Q4	Q5
$\Delta ln(earnings), yearly$	0.339***	0.354***	0.253***	0.242***	0.178**
	(0.086)	(0.086)	(0.084)	(0.067)	(0.075)
N Individuals	197,439	231,659	233,591	221,801	187,853
N Total	1,356,504	1,446,339	1,363,573	1,321,365	1,357,949

Notes: The Table displays estimates for τ in specification 1, estimated separately by subsample on the outcomes displayed in column 1. The subsamples are defined based on quintiles of the distribution of employment concentration in an employee's local labor market, defined as county $\times$ sector. Each specification includes individual and county fixed effects, as well as year-by-sector by within-sector firm-size tercile fixed effects. The models additionally allow yearly fixed effects to vary over: quintiles of the energy intensity of local production, measured as the share of establishments in each county that are operating in energy-intensive sectors (i.e., agriculture, utilities, transportation, manufacturing), as well as across differences in the lagged county-level unemployment rate. Standard errors are clustered at the county level and displayed in parentheses.

occupations and sectors. If fewer such outside options are available, or if a person lacks sufficiently general skills to take advantage of such switches, we should naturally expect a lower return to E-E mobility. In line with this, Table A11 shows a lower rate of pass-through in local labor markets with higher levels of employment concentration.

The generation of outside offers not only determines opportunities for job-to-job mobility but also the bargaining power of workers inside their firm. Keeping the two channels separate helps distinguish between workers with and without a university degree. Table A12 demonstrates that in response to energy price increases, relatively lower-skilled workers respond at the job-to-job mobility margin. This allows them to keep the gap in average pass-through relative to their university-educated peers comparatively limited. The latter group seems better able to negotiate wage increases while staying on the job.

While the gap in average earnings adjustments between the two groups is not enormous, note that households in which the household head has a university degree typically spend less of their overall budget on energy, even conditional on income and other observable characteristics (see Appendix A.3 for details). This implies that the energy cost instrument is likely to overestimate cost increases for this subgroup of workers, suggesting that they are de facto able to recover an even higher share than the estimated 35% within a year.

Table A12: The Effect of Cost Shocks on Earnings and E-E, by Education

	Pr(E-E)	$\Delta(earnings)$	$\Delta(earnings)\|staying$	$\Delta(earnings)\|switching$
Sample:				
Non-University	0.298*	0.281***	0.212***	0.499***
	(0.157)	(0.042)	(0.032)	(0.128)
University	0.167	0.347***	0.364***	0.637**
	(0.225)	(0.113)	(0.083)	(0.275)

Notes: The Table displays estimates for τ in specification 1, estimated separately by subsample on the outcomes displayed in row 1. Each specification includes individual and county fixed effects, as well as year-by-sector by within-sector firm-size tercile fixed effects. The models additionally allow yearly fixed effects to vary over: quintiles of the energy intensity of local production, measured as the share of establishments in each county that are operating in energy-intensive sectors (i.e., agriculture, utilities, transportation, manufacturing), as well as across differences in the lagged county-level unemployment rate. Standard errors are clustered at the county level and displayed in parentheses.

A.5 Theoretical Appendix

A.5.1 Proofs.

In the following I prove the claim made in Section 5.2 that, for a given set of wages, the employment-weighted average elasticity of labor supply increases in q, as long as $F(\beta)$ has finite first and second moments, there are no superstar firms, and the dispersion in market wages is not too extreme. I consider the case in which all firm wages, as well as unemployment benefits, are strictly above the subsistence level of energy expenditures before and after price changes. Since energy expenditure shares are typically less than 10%, this covers all empirically relevant cases of energy price changes.

For notational convenience, define the following objects for each firm j in a given local labor market:

$$\phi_j(q) = \frac{w_j}{w_j - q\bar{e}}, \qquad x_j(q) = \ln\big(w_j - q\bar{e}\big), \quad B(q) = \sum_{j=1}^{J} Pr(j) = 1 - Pr(0),$$
$$N_j(q) = \int \beta\, Pr(j|\beta)\big(1 - Pr(j|\beta)\big)\, dF(\beta).$$

The employment-weighted labor-supply elasticity is $\bar{\varepsilon}(q) = \dfrac{A(q)}{B(q)}$ with $A(q) = \sum_{j=1}^{J} \phi_j(q)\, N_j(q)$.

For later bounds, also define:

$$\begin{aligned}
w_{\min} &= \min_j w_j, \quad w_{\max} = \max_j w_j, \\
\phi'_{\min} &\equiv \min_{j,q} \frac{\partial \phi_j}{\partial q} = \frac{w_{\max}\bar{e}}{\big(w_{\max} - q_{\min}\bar{e}\big)^2}, \quad \phi_{\max} \equiv \max_{j,q} \phi_j(q) = \frac{w_{\min}}{w_{\min} - q_{\max}\bar{e}} \\
\Delta'_x(\kappa) &\equiv \max_j x'_j(q) - \min_j x'_j(q) = \bar{e}\left[\frac{1}{w_{\min} - q_{\max}\bar{e}} - \frac{1}{\kappa\, w_{\min} - q_{\max}\bar{e}}\right] \\
\Delta'_{\text{all}}(\kappa) &\equiv \max_{j\in\{0,1,\ldots,J\}} x'_j(q) - \min_{j\in\{0,1,\ldots,J\}} x'_j(q) = \bar{e}\left[\frac{1}{b - q_{\max}\bar{e}} - \frac{1}{\kappa\, w_{\min} - q_{\max}\bar{e}}\right],
\end{aligned}$$

where $\kappa \equiv \kappa_w = w_{\max}/w_{\min} > 1$ describes the dispersion of wages in the local labor market. Note that $\Delta'_x(\kappa)$ relates to a across-firm comparison, whereas $\Delta'_{all}(\kappa)$ also includes the outside option of non-employment.

Using this notation, the following proposition summarizes the discussed relationship between the employment-weighted average elasticity of labor supply and energy prices q:

Proposition 1. *Fix a local market, wages $(w_j)_{j\geq 1}$, unemployment benefits b, and an interval of energy prices $q \in q_0 = [q_{\min}, q_{\max}]$. Assume:*

- ***A1: Energy price increases are not too extreme:*** *For all j and all $q \in q_0$, $w_j - q\bar{e} > 0$ and $b - q\bar{e} > 0$.*

- ***A2: No superstar firms:*** *For all $j \geq 1$ and all β, $Pr(j|\beta) \leq \frac{1}{2}$ for all $q \in q_0$.*

- ***A3: Non-trivial employment mass on a high-β set:*** *There exist $\alpha \in (0,1)$, its quantile β_α, the set $G_\alpha = \{\beta \geq \beta_\alpha\}$ with $F(G_\alpha) = 1 - \alpha$, and $\theta_0 \in (0,1)$ such that for all $\beta \in G_\alpha$ and all $q \in q_0$, $\sum_{j=1}^{J} Pr(j|\beta) \geq \theta_0$. Let $B_{\min} \equiv \theta_0(1-\alpha) > 0$.*

- ***A4: Regularity of*** *$F(\beta) : \mathbb{E}[\beta] < \infty$ and $\mathbb{E}[\beta^2] < \infty$.*

- ***A5: Explicit dispersion cap on*** *κ_w: The within-market wage dispersion ratio $\kappa_w = w_{\max}/w_{\min}$ satisfies $\kappa_w \leq \bar{\kappa}_w$, where $\bar{\kappa}_w > 1$ is the value solving*

$$\frac{1}{2}\,\beta_\alpha\, B_{\min}\,\phi'_{\min} \;=\; \underbrace{\frac{1}{2}\,\phi_{\max}\,\Delta'_x(\kappa)\,\mathbb{E}[\beta^2]}_{c_{\text{sort}}(\kappa)} \;+\; \underbrace{\frac{\phi_{\max}}{B_{\min}}\cdot\frac{J}{4}\,\mathbb{E}[\beta]^2\,\Delta'_{\text{all}}(\kappa)}_{B_{\min}\,\bar{C}_{\text{unemp}}(\kappa)}\,. \tag{12}$$

Under Assumptions A1–A5, for all $q \in q_0$,

$$\frac{\partial \bar{\varepsilon}(q)}{\partial q} \;>\; 0.$$

Proof. Write $\bar{\varepsilon}(q) = A(q)/B(q)$ with $A(q) = \sum_j \phi_j(q) N_j(q)$ and $B(q) = 1 - Pr(0)$. Differentiating with respect to q yields:

$$\bar{\varepsilon}'(q) = \frac{A'(q)}{B(q)} + \frac{A(q)\,Pr'(0)}{B(q)^2}. \tag{13}$$

Note that $B(q)$ measures the share of employed workers, which in the given scenario is always positive. We can thus focus on the numerators.

Step 1: Bound $A'(q)$. Decompose

$$A'(q) = \sum_j \phi'_j(q) N_j(q) \;+\; \underbrace{\sum_j \phi_j(q) N'_j(q)}_{\mathcal{T}(q)}.$$

By Assumptions A2 and A3,

$$\sum_j N_j(q) = \sum_j \int \beta\, Pr(j)(1 - Pr(j))\, dF \;\geq\; \tfrac{1}{2}\,\beta_\alpha \int_{G_\alpha} \sum_j Pr(j)\, dF \;\geq\; \tfrac{1}{2}\,\beta_\alpha\, B_{\min}.$$

Hence, the first term of $A'(q)$ satisfies,

$$\sum_j \phi_j'(q) N_j(q) \;\geq\; \tfrac{1}{2}\,\beta_\alpha\, B_{\min}\, \phi'_{\min}. \tag{14}$$

For the second term, $\mathcal{T}(q)$, note that:

$$\begin{aligned}
N_j'(q) &= \int \beta \left[Pr'(j|\beta)(1-2Pr(j|\beta))\right] dF(\beta) \\
&= \int \beta^2\, Pr(j|\beta)(1-2Pr(j|\beta)) \Big[x_j'(q) - \sum_k Pr(k|\beta)\, x_k'(q)\Big]\, dF(\beta) \\
\mathcal{T}(q) &= \sum_j \phi_j(q) \int \beta^2\, Pr(j|\beta)(1-2Pr(j|\beta)) \Big[x_j'(q) - \sum_k Pr(k|\beta)\, x_k'(q)\Big]\, dF(\beta) \\
|\mathcal{T}(q)| &\leq \sum_j \phi_j(q) \int \beta^2\, Pr(j|\beta)|1-2Pr(j|\beta)| \Big|\Big[x_j'(q) - \sum_k Pr(k|\beta)\, x_k'(q)\Big]\Big|\, dF(\beta),
\end{aligned}$$

where the last line follows from an application of the triangle inequality.

By Assumption A2, $Pr(j|\beta)|1-2Pr(j|\beta)| \leq 1/2$. Also $\phi_j \leq \phi_{max}$ and $\sum_j Pr(j|\beta) < 1$ and thus:

$$\begin{aligned}
|\mathcal{T}(q)| &\leq \frac{1}{2}\phi_{max} \int \beta^2 \sup_j \Big|\Big[x_j'(q) - \sum_k Pr(k|\beta)\, x_k'(q)\Big]\Big| dF(\beta) \\
&= \frac{1}{2}\phi_{max} \int \beta^2 (\max_j x_j'(q) - min_j x_j'(q)) dF(\beta) \\
&= \frac{1}{2}\phi_{max} \mathbb{E}[\beta^2] (\max_j x_j'(q) - min_j x_j'(q)),
\end{aligned}$$

since the second moment of β exists by Assumption A4. As $(\max_j x_j'(q) - min_j x_j'(q))$ is increasing in q, we know that:

$$|\mathcal{T}(q)| \leq \frac{1}{2}\phi_{max}\mathbb{E}[\beta^2]\, \underbrace{(\max_j x_j'(q_{max}) - min_j x_j'(q_{max}))}_{=\Delta_x'(\kappa)} = \underbrace{\frac{1}{2}\phi_{max}\mathbb{E}[\beta^2]\Delta_x'(\kappa)}_{c_{sort}(\kappa)}. \tag{15}$$

Combining (14) and (15):

$$A'(q) \;\geq\; \frac{1}{2}\beta_\alpha B_{min}\phi'_{min} - c_{sort}(\kappa) \tag{16}$$

Step 2: Bound $A(q)Pr'(0)$, i.e., changes in unemployment. Since $Pr(j)(1 - Pr(j)) \leq \frac{1}{4}$ (by assumption A2) and since $\mathbb{E}[\beta]$ is finite by A4,

$$A(q) \leq \phi_{\max} \frac{J}{4} \mathbb{E}[\beta].$$

Moreover:

$$Pr'(0) = \int \beta Pr'(0|\beta) dF(\beta) = \int \beta Pr(0|\beta)(x_0' - \sum_{k \geq 0} Pr(k|\beta) x_k') dF(\beta)$$
$$|Pr'(0)| \leq \Delta_{all}'(q)\mathbb{E}[\beta] \leq \Delta_{all}'(\kappa_w)\mathbb{E}[\beta],$$

where the second line follows the same logic as above and the definition of $\Delta_{\text{all}}'(\kappa) \equiv \max_{j \in \{0,1,\ldots,J\}} x_j'(q) - \min_{j \in \{0,1,\ldots,J\}} x_j'(q)$. Note that if $b < w_{min}$, which is the empirically relevant case, then $Pr'(0) < 0$.

Using $B(q) \geq B_{\min}$, we can collect terms into:

$$\left| \frac{A(q)\, Pr'(0)}{B(q)^2} \right| \leq \frac{\phi_{\max}}{B_{\min}^2} \cdot \frac{J}{4} \mathbb{E}[\beta]^2 \, \Delta_{\text{all}}'(\kappa_w) \equiv \bar{C}_{\text{unemp}}(\kappa_w). \tag{17}$$

Step 3: Combine the Elements. Inserting (16), and (17) into the derivative of the employment-weighted labor supply elasticity, we get:

$$\bar{\varepsilon}'(q) \geq \frac{\frac{1}{2}\beta_\alpha B_{min} \phi_{min}' - c_{sort}(\kappa)}{B_{min}} - \bar{C}_{unemp}(\kappa_w).$$

By Assumption A5, this is strictly larger than zero as the right-hand side of equation (12) is strictly increasing in κ and $\kappa_w < \bar{\kappa}_w$. Hence $\bar{\varepsilon}'(q) > 0$ for all $q \in q_0$. □

The proposition establishes a bound on wage dispersion under which the employment-weighted labor supply elasticity increases monotonically with q. To provide some intuition on the bound, consider that the derivative of the labor supply elasticity contains a mechanically increasing part in

$$\sum_j \phi_j'(q) N_j(q) = \sum_j \frac{w_j \bar{e}}{(w_j - q\bar{e})^2} N_j(q),$$

which captures the fact that workers' marginal utility of consumption rises as energy prices rise. The bound scales the minimum of this effect by a non-trivial mass of high-β workers ($\frac{1}{2}\beta_\alpha B_{min}$). As long as there is a mass of employed workers who care deeply about consumption, this mechanical relationship has bite. At the same time, workers switch in response to energy cost shocks towards

higher-paying firms. This changes the distribution of labor supply elasticities across firms. It is not necessarily the case that higher-paying firms have higher labor supply elasticities. If wages aren't too dispersed, the resorting across firms is dampened, and potential counteracting forces from resorting are limited.

Importantly, this bound is a sufficient (conservative) condition but not necessary. The following corollary further shows that the elasticity of labor supply is more likely to increase in q in markets where consumers are more exposed to the cost shock (i.e., higher $\bar{e}$). For this, we need only to sharpen Assumptions A1 and A3 slightly:

A1′: Energy price increases are not too extreme: For all j and all $q \in q_0$, $w_j - q\bar{e} > 0$ and $b - q\bar{e} > 0$. Also $\frac{q_{max}\bar{e}}{w_{min}} < \frac{1}{2} \forall \bar{e}$.

A3′: Non-trivial employment mass on a high-β set is uniform across q: There exist $\alpha \in (0,1)$, its quantile β_α, and $\theta_0 \in (0,1)$ such that for all $q \in [q_{\min}, q_{\max}]$ and all $\beta \geq \beta_\alpha$, $\sum_{j\geq 1} \Pr(j \mid \beta) \geq \theta_0$. Let $B_{\min} \equiv \theta_0(1-\alpha) > 0$.

Assumption A1 is revised to allow a maximum energy expenditure share of 50% even in high-expenditure markets and among low-income individuals. Note that in the highest expenditure decile, energy still accounts for only around 13%, so this assumption covers all empirically relevant cases. The difference relative to Assumption A3 is mainly technical in nature and requires the existence of a minimum B_{min} that is valid for any q. This rules out limit cases, in which the mass of high-β workers converges to zero. With this, the following corollary holds:

Corollary 1. *Maintain Assumptions A2 and A4 of Proposition 1, and replace A1 with A1′ and A3 with A3′. For each $\bar{e} > 0$, define $\bar{\kappa}_w(\bar{e}) > 1$ as the unique solution in κ such that equation (12) holds with equality:*

$$\frac{1}{2}\,\beta_\alpha B_{\min}\,\phi'_{\min}(\bar{e}) = \frac{1}{2}\,\mathbb{E}[\beta^2]\,\phi_{\max}(\bar{e})\,\Delta'_x(\kappa) + \frac{J}{4B_{\min}}\,\mathbb{E}[\beta]^2\,\phi_{\max}(\bar{e})\,\Delta'_{\text{all}}(\kappa).$$

Then $\bar{\kappa}_w(\bar{e})$ is strictly increasing in $\bar{e}$. Consequently, a higher $\bar{e}$ implies that larger dispersions of wages are admissible for the sufficient condition in Proposition 1 to be satisfied.

Proof. Define

$$F(\bar{e},\bar{\kappa}) = \frac{1}{2}\,\beta_\alpha B_{\min}\,\phi'_{\min}(\bar{e}) - \mathbb{E}[\beta^2]\phi_{max}(\bar{e})\left(\frac{1}{2}\,\Delta'_x(\bar{\kappa}) + \frac{J}{4B_{\min}}\,\Delta'_{\text{all}}(\bar{\kappa})\right) = 0.$$

Then, by the implicit function theorem,

$$\frac{\partial \bar{\kappa}}{\partial \bar{e}} = -\frac{F_e(\bar{e}, \bar{\kappa})}{F_\kappa(\bar{e}, \bar{\kappa})}.$$

The derivative with respect to $\bar{\kappa}$ is:

$$F_\kappa(\bar{e}, \bar{\kappa}) = -\mathbb{E}[\beta^2]\phi_{max}(\bar{e}) \left[\frac{1}{2}\frac{\partial \Delta'_x(\bar{\kappa})}{\partial \bar{\kappa}} + \frac{J}{4B_{min}}\frac{\partial \Delta'_{all}(\bar{\kappa})}{\partial \bar{\kappa}}\right].$$

By A1′, $\phi_{max}(\bar{e}) > 0$ and by A3′, $B_{min} > 0$. The two spreads $\Delta'_x(\kappa)$ and $\Delta'_{all}(\kappa)$ are by definition increasing in κ. It thus follows that $F_\kappa(\bar{e}, \bar{\kappa}) < 0$, such that the sign of $\partial\bar{\kappa}/\partial\bar{e}$ depends on $F_{\bar{e}}(\bar{e}, \bar{\kappa})$.

The derivate with respect to $\bar{e}$ is:

$$\begin{aligned}
F_{\bar{e}}(\bar{e}, \bar{\kappa}) &= \frac{1}{2}\beta_\alpha B_{min}\frac{\partial \phi'_{min}(\bar{e})}{\partial \bar{e}} - \frac{\partial \phi_{max}(\bar{e})}{\partial \bar{e}}\mathbb{E}[\beta^2]\left(\frac{1}{2}\ \Delta'_x(\bar{\kappa}) + \frac{J}{4B_{\text{min}}}\ \Delta'_{\text{all}}(\bar{\kappa})\right) \\
&= \frac{1}{2}\beta_\alpha B_{min}\frac{\partial \phi'_{min}(\bar{e})}{\partial \bar{e}} - \frac{\partial \phi_{max}(\bar{e})}{\partial \bar{e}}\left[\frac{\frac{1}{2}\,\beta_\alpha B_{\text{min}}\,\phi'_{\text{min}}(\bar{e})}{\phi_{max}(\bar{e})}\right] \\
&= \frac{1}{2}\beta_\alpha B_{min}\phi'_{min}(\bar{e})\left[\frac{\partial \phi'_{min}(\bar{e})/\partial \bar{e}}{\phi'_{min}(\bar{e})} - \frac{\partial \phi_{max}(\bar{e})/\partial \bar{e}}{\phi_{max}(\bar{e})}\right] \\
&= \frac{1}{2}\beta_\alpha B_{min}\phi'_{min}(\bar{e})\left[\frac{\partial ln(\phi'_{min}(\bar{e}))}{\partial \bar{e}} - \frac{\partial ln(\phi_{max}(\bar{e}))}{\partial \bar{e}}\right],
\end{aligned}$$

where the second equality follows from substituting terms at $F(\bar{e}, \bar{\kappa}) = 0$.

By A1′ and A3′, $\frac{1}{2}\beta_\alpha B_{min}\phi'_{min}(\bar{e}) > 0$. Computing the derivatives inside the brackets and substituting in yields:

$$F_{\bar{e}}(\bar{e}, \bar{\kappa}) = \frac{1}{2}\beta_\alpha B_{min}\phi'_{min}(\bar{e})\left[\frac{1}{\bar{e}} + \frac{2q_{min}}{w_{max} - q_{min}\bar{e}} - \frac{q_{max}}{w_{min} - q_{max}\bar{e}}\right].$$

By A1′, the denominators are positive. A sufficient condition for $F_{\bar{e}}(\bar{e}, \bar{\kappa}) > 0$ is thus

$$\frac{1}{\bar{e}} > \frac{q_{max}}{w_{min} - q_{max}\bar{e}} \iff \frac{q_{max}\bar{e}}{w_{min}} < \frac{1}{2}.$$

This inequality holds by Assumption A1′ and thus $F_{\bar{e}}(\bar{e}, \bar{\kappa}) > 0 \implies \frac{\partial \bar{\kappa}}{\partial \bar{e}} > 0$.

The set of admissible wage dispersion levels that support a monotonic increase in the weighted labor supply elasticity after energy price increases is thus expanding in $\bar{e}$. □

A.5.2 Additional Details on the Estimation of the Baseline Framework

This section provides details on the external calibration of preference parameters in workers' utility functions over consumption, the construction of the worker panel, model fit, and the firm's optimization problem.

Calibration of $\mu, \bar{e}_l$. In the model, workers' preferences for consumption are described by:

$$U(c, e|\bar{e}_l) = \mu ln(c) + (1-\mu) ln(e - \bar{e}_l).$$

With income y_j and energy price q, this yields the following optimal demand and energy expenditure share:

$$(c^*, e^*) = \left(\mu(y_j - q\bar{e}_l), (1-\mu)\frac{y_j - q\bar{e}_l}{q} + \bar{e}_l\right)$$
$$S_i^e = (1-\mu) + \mu\frac{q\bar{e}_l}{y_{j(i)}}.$$

Let the average energy expenditure share in a labor market be denoted by S_l^e. For the empirical analyses in the paper, I use only residual geographic variation in S_l^e after adjusting for socio-demographic worker sorting. To replicate this in the theoretical part, the only variation in average expenditure shares across local labor markets comes from $\bar{e}$. Fixing $\bar{y}$ and letting $q = 1$ in the baseline period of the model:

$$\bar{e}_l = \frac{\bar{y}}{\mu}\left(S_l^e - (1-\mu)\right). \tag{18}$$

Thus, for a known μ, $\bar{e}_l$ can be directly inferred from the data by replacing S_l^e in (18) with the sample analogue. In my case, this is proxied by $\sum_g S_c^g$, i.e., the sum over all energy types g, where c refers to a county - the local labor market proxy in my main analysis.

To calibrate μ, consider the mean value of S across local labor markets and the derivative with respect to y at that mean:

$$\bar{S} = (1-\mu) + \frac{\mu}{\bar{y}}\underbrace{\frac{1}{L}\sum_l \bar{e}_l}_{\equiv\bar{\bar{e}}}$$
$$\bar{S}' = -\frac{\mu\bar{\bar{e}}}{\bar{y}^2}.$$

Let $\beta_{S,y}$ be the coefficient obtained from a regression of S on y, then to a first-order approximation, we can write that $\bar{S}' \approx \beta_{S,y} \iff \mu \frac{\bar{\bar{e}}}{\bar{y}} \approx -\beta_{S,y}\bar{y}$. It is thus possible to write the mean of S as:

$$\bar{S} = (1-\mu) + \mu \frac{\bar{\bar{e}}}{\bar{y}} \approx (1-\mu) - \beta_{S,y}\bar{y}$$
$$\iff \mu \approx 1 - \bar{S} - \beta_{S,y}\bar{y}.$$

Using the sample analog for $\bar{S}$ and $\bar{y}$, as well as the regression coefficient on the relationship between total expenditures and energy expenditure shares (see Section A.3), yields a calibrated $\mu = 0.951$. With this information, equation (18) allows for a direct calibration of arbitrarily many local minimum consumption levels $\bar{e}$.

Construction of the Worker Panel. The reduced-form empirical approach of this paper relies on a comparison of the response in job transitions and earnings to energy price changes across labor markets with a different baseline energy expenditure share but otherwise similar make-up. To replicate this idea within my theoretical framework, I construct a panel of workers across 10 stylized labor markets that represent deciles of the energy-expenditure share distribution. Based on the calibrated μ above, the corresponding values for $\bar{e}$ are:

	Energy Expenditure Decile:									
	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
S_{Data}	0.051	0.055	0.060	0.066	0.069	0.071	0.074	0.077	0.081	0.088
$\bar{e}$	0.266	0.717	1.259	1.880	2.196	2.478	2.760	3.133	3.607	4.307

Apart from the variation in $\bar{e}$, all labor markets contain the exact same set of 54 firms in terms of productivity and amenities, which are drawn once and then reused across the stylized markets. The number of firms is chosen in line with the average number of establishments per county in the main empirical data set. The data show that each firm employs an average of 18 workers, and the average unemployment rate over the estimation window is 7.4%. I consequently simulate $N = \frac{18*54}{1-0.074} \approx 1050$ workers per labor market in the panel.

Since the model is static, I simulate a panel by solving the labor market equilibrium at different values of q and tracking workers' choices. To simplify the calculation, I treat workers as continuous when solving the labor market equilibrium in each market. That is, I assume there is a mass of $N = 1$ workers in each labor market, and firms set wages according to this continuum of workers. The estimation routine follows the following logic:

1. Guess a parameter vector Θ. Based on this guess, draw firm productivities, amenities, and energy intensities for a set of 54 firms, as well as workers' relative tastes for consumption.

2. For each local labor market l, simulate a baseline period in which $q = 1$ and solve for the labor market equilibrium.

3. Based on the equilibrium wages from the previous step and the realized taste shocks of workers, assign the discrete set of workers to firms.

4. Interpret the baseline period as a steady state and calculate the following static moments across all local labor markets: the mean of wages, the variance of wages, the unemployment rate, and the variance of firm-level employment counts. At the firm level, also estimate a linear cross-sectional regression of employment count on wages and use it as an additional moment.

5. Based on the mean wage across all labor markets and labor-market specific $\bar{e}$, calculate $\bar{S}_l$.

6. Consider a set $\tilde{q} = \{0.87; 1.09; 1.20; 1.25; 1.36\}$, representing quintile-averages of the empirically observed relative changes in average (expenditure-weighted) energy prices over five-year windows. Note that, since $q = 1$ in period 1, any $q' \in \tilde{q}$ is not only the relative price change but also the price of energy in period 2. For each $q' \in \tilde{q}$:

   - Solve the labor market equilibrium in each local labor market and save firms' wages.
   - Since the model does not contain mechanisms that generate switching without changes in energy prices, allow a fraction π to redraw their idiosyncratic tastes in the second period. Calibrate π to match average switching rates.
   - Re-allocate workers to firms based on their preference draws and the new wages. Track if their firm choice changed.
   - For each worker, construct a regressor: $C_{l(i)} = \bar{S}_{l(i)} * q'$ based on the baseline period expenditure share $\bar{S}_{l(i)}$.

7. For each period-to-period comparison, keep the set of workers who chose to be employed in both periods. Stack all observations into a panel and estimate regressions of the type $y_i = \alpha_i + \delta_t + \tau_1 C_{l(i)} + u_i$, where y_i is either an indicator for switching firms or the change in log wages, and δ_t refers to different draws from $\tilde{q}$. Note that because workers are tied to a labor market in the model, the simulated regression has no labor market fixed effects. Since

the model has no sectors, the regression also does not contain sector fixed effects, contrary to the empirical regression.

8. Repeat the estimation in the previous bullet point, conditional on workers' switching patterns between periods.

9. Construct the dynamic moments by matching the model coefficients to their data counterparts.

10. Let $M(\Theta)$ be the vector of all simulated static and dynamic moments, and let M^D be the empirical counterpart. Construct the following objective:

$$\mathcal{T} = (M(\Theta) - M^D)' * \mathcal{I} * (M(\Theta) - M^D),$$

where $\mathcal{I}$ is the identity matrix.

11. Repeat steps 1 to 10 to minimize $\mathcal{T}$.

Model Fit. Estimating the model yields the parameters displayed in Table 8. The fit between data moments and model-implied moments is summarized in the following table:

Table A13: Moment Fit of Estimated Parameters

Data Moments	M^D	$M(\Theta^*)$
Reg. Coeff. of Δw on $S_c \frac{q'}{q}$	0.804	0.780
Reg. Coeff. of Δw on $S_c \frac{q'}{q}$ | switching	1.258	1.253
Reg. Coeff. of Δw on $S_c \frac{q'}{q}$ | staying	0.398	0.393
Reg. Coeff. of $Pr(E-E)$ on $S_c \frac{q'}{q}$	0.330	0.319
Mean(wage)	107.34	103.26
ln Var(wage)	6.969	7.372
Unemployment Rate	0.074	0.076
Firm Size P75/P25	6.40	11.07
Wage P90/P50	1.612	1.703

Notes: M^D denotes the empirically observed moment, whereas $M(\theta^*)$ denotes the simulated moments at the optimal parameter value.

The model successfully reproduces the mean and variance of wages, the unemployment rate, and the pass-through of energy prices into earnings, both at the average and within the sample of switchers and stayers, respectively. It also matches the job-to-job mobility responses to energy cost shocks. The model fit is considerably worse at matching the interquartile range of firm-level

employment, generating substantial excess dispersion. Given that the only modeled reason for firm-level size dispersion besides wages is captured by the distribution of average non-wage attributes σ_{ξ}, and there are no further frictions to firm-choice, this is partially a structural issue.

The Firm's Decision Problem and the Labor Market Equilibrium. For estimation, I impose that firms produce with a Cobb-Douglas technology such that their maximization problem is:

$$\max_{w_j, E_j}\{\Pi_j = z_j \left[L_j^{1-\gamma_j} E_j^{\gamma_j}\right]^{\alpha} - qE_j - w_j L_j \quad s.t. \quad L_j(w_j|\mathbf{W}, q) = N \times Pr(j|\mathbf{W}, q), \tag{19}$$

where, for simplicity, it is assumed that firms and consumers pay the same price q for energy. From here, the first order conditions are:

$$\begin{aligned}
& E_j : z_j L_j^{(1-\gamma_j)\alpha} E_j^{\alpha\gamma_j - 1} \alpha\gamma_j - q = 0 \\
\iff & E_j = \left[\frac{\alpha\gamma_j}{q} z_j L_j^{(1-\gamma_j)\alpha}\right]^{\frac{1}{1-\alpha\gamma_j}} \qquad (20) \\
& L_j : z_j \alpha(1-\gamma_j) L_j^{(1-\gamma_j)\alpha - 1} E_j^{\alpha\gamma_j} \frac{\partial L_j}{\partial w_j} - L_j - w_j \frac{\partial L_j}{\partial w_j} = 0 \\
\iff & z_j \alpha(1-\gamma_j) L_j^{(1-\gamma_j)\alpha - 1} E_j^{\alpha\gamma_j} \underbrace{\frac{\partial L_j}{\partial w_j}\frac{w_j}{L_j}}_{=\epsilon_j} - w_j - w_j \underbrace{\frac{\partial L_j}{\partial w_j}\frac{w_j}{L_j}}_{=\epsilon_j} = 0 \\
\iff & z_j \alpha(1-\gamma_j) L_j^{(1-\gamma_j)\alpha - 1} E_j^{\alpha\gamma_j} \frac{\epsilon_j}{1+\epsilon_j} = w_j. \qquad (21)
\end{aligned}$$

Inserting (20) into (21) yields:

$$\begin{aligned}
w_j &= z_j \alpha(1-\gamma_j) L_j^{(1-\gamma_j)\alpha - 1} \left[\frac{\alpha\gamma_j}{q} z_j L_j^{(1-\gamma_j)\alpha}\right]^{\frac{\alpha\gamma_j}{1-\alpha\gamma_j}} \frac{\epsilon_j}{1+\epsilon_j} \\
&= \alpha(1-\gamma_j)\left(\frac{\alpha\gamma_j}{q}\right)^{\frac{\alpha\gamma_j}{1-\alpha\gamma_j}} \left(\frac{z_j}{L_j^{1-\alpha}}\right)^{\frac{1}{1-\alpha\gamma_j}} \frac{\epsilon_j}{1+\epsilon_j}. \qquad (22)
\end{aligned}$$

Firms own-wage labor supply elasticity is derived in Section 5 and equals:

$$\epsilon_j = \frac{w_j}{Pr(j)} \int \frac{\partial Pr(j|\beta)}{\partial w_j} dF(\beta) = \frac{w_j}{w_j - q\bar{e}} \underbrace{\int \beta \times \frac{Pr(j|\beta)(1 - Pr(j|\beta))}{Pr(j)} dF(\beta)}_{\equiv \beta_j}, \tag{23}$$

where:

$$Pr(j|\beta) = \frac{exp(\beta ln(w_j - q\bar{e}) + \bar{\xi}_j)}{\sum_{j'} exp(\beta ln(w_{j'} - q\bar{e}) + \bar{\xi}_{j'})}. \tag{24}$$

Using (22) to (24) jointly for all firms j in a local labor market, yields a fixed point problem of wages, in which firms best respond to the posted wages of other firms. To solve this problem, the assumption of atomistic firms is not necessary. In fact, the labor supply elasticity of each firm is a function of its size, reflecting an oligopsony component in addition to the market power firms derive from heterogeneity in worker tastes (Berger, Herkenhoff and Mongey, 2022).

Chan et al. (2024) prove that with decreasing returns to scale ($\alpha < 1$) and extreme value taste shocks (ξ_{ij}), a labor market equilibrium exists and is unique. They also provide simple iterative algorithms that converge to the equilibrium. In my setting, the complication is that β is not a fixed scalar but a random variable, so unconditional choice probabilities and elasticities generally lack closed-form expressions. I therefore adapt their algorithm by approximating the log-normal distribution of β with Gauss–Hermite quadrature (GHQ). In practice, I use $n = 7$ GHQ nodes $\{t_k, w_k^{\text{GH}}\}_{k=1}^{n}$ from Matlab's numerical library, and map them into lognormal draws $\beta_k = \exp\left(\mu_\beta + \sqrt{2}\,\sigma_\beta\, t_k\right)$ and weights $\omega_k = \frac{w_k^{\text{GH}}}{\sqrt{\pi}}$. This allows me to approximate any integral with respect to $F(\beta)$ by:

$$\int g(\beta)\, dF(\beta) \;\approx\; \sum_{k=1}^{n} \omega_k\, g(\beta_k).$$

Using this approximation, I otherwise proceed in the standard way:

1. Guess an initial vector $\mathbf{W_0}$.
2. Compute workers' choice probabilities and firms' marginal revenue products of labor.
3. For each firm, calculate the best wage-response given the vector of wages.
4. Update $\mathbf{W_0}$ with the best responses and re-initialize the procedure until convergence.